CHAPTER 11

Spectroscopy

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: a3cf72bf7daba7d76cb851eaad5e894d929974d40d2ea1bb0feaf8649082f2fe

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch11:exercise:ir-read | ir-read:stage-1, ir-read:stage-2, ir-read:stage-3

ch11:exercise:uv-energy | uv-energy:stage-1, uv-energy:stage-2, uv-energy:stage-3

ch11:exercise:nmr-ethyl | nmr-ethyl:stage-1, nmr-ethyl:stage-2, nmr-ethyl:stage-3

ch11:exercise:combined-data | combined-data:stage-1, combined-data:stage-2, combined-data:stage-3

ch11:lesson:ir | ir:rule, ir:cue, ir:trap, ir:app, ir:step-1, ir:step-2, ir:step-3

ch11:lesson:uv | uv:rule, uv:cue, uv:trap, uv:app, uv:step-1, uv:step-2

ch11:lesson:nmr | nmr:rule, nmr:cue, nmr:trap, nmr:app, nmr:step-1, nmr:step-2, nmr:step-3

ch11:lesson:combine | combine:rule, combine:cue, combine:trap, combine:app, combine:step-1, combine:step-2, combine:step-3

ch11:molecule:acetone | acetone

ch11:molecule:isopropanol | isopropanol

ch11:molecule:aceticacid | aceticacid

ch11:molecule:butadiene | butadiene

ch11:molecule:ethene | ethene

ch11:molecule:ethylacetate | ethylacetate

ch11:molecule:acetaldehyde | acetaldehyde

ch11:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 11.1 Infrared Spectroscopy

scope-2: 11.2 Ultraviolet Spectroscopy

scope-3: 11.3 Nuclear Magnetic Resonance Spectroscopy

Objective: Recognize approximate IR regions

Objective: Relate conjugation, photon energy, and absorbance

Objective: Interpret NMR shifts, integrals, and splitting

Supplied Kaplan curriculum: Chapter 11; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter11b.xhtml

IR functional-group regions: OpenStax 12.8 | https://openstax.org/books/organic-chemistry/pages/12-8-infrared-spectra-of-some-common-functional-groups

First-order proton splitting: OpenStax 13.6 | https://openstax.org/books/organic-chemistry/pages/13-6-spin-spin-splitting-in-1h-nmr-spectra

Conjugation and UV absorption: OpenStax 14.8 | https://openstax.org/books/organic-chemistry/pages/14-8-interpreting-ultraviolet-spectra-the-effect-of-conjugation

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

ir | IR: recognize functional groups

Vibrations must change dipole moment

ir:rule | Core Rules

IR probes molecular vibrations. A vibration absorbs IR when it changes the molecular dipole moment. Useful approximate regions: alcohol O–H 3200–3600 cm^{-1} (often broad); acid O–H 2500–3300 (very broad); C=O about 1650–1800; nitrile about 2210–2260. N–H is often near 3300, generally less broad than hydrogen-bonded OH.

ir:cue | Memory Cues

Use a band pattern, not one magic number. The fingerprint region below about 1500 cm^{-1} helps comparison.

ir:trap | Common Mistakes / MCAT Traps

Carbonyl positions vary with functional group, conjugation, ring strain, and environment. Absence of an OH band may support a structure, but sample conditions matter. IR alone seldom gives complete connectivity.

ir:app | MCAT Applications

Follow loss of a carbonyl band and appearance of OH when monitoring a carbonyl reduction, together with other evidence.

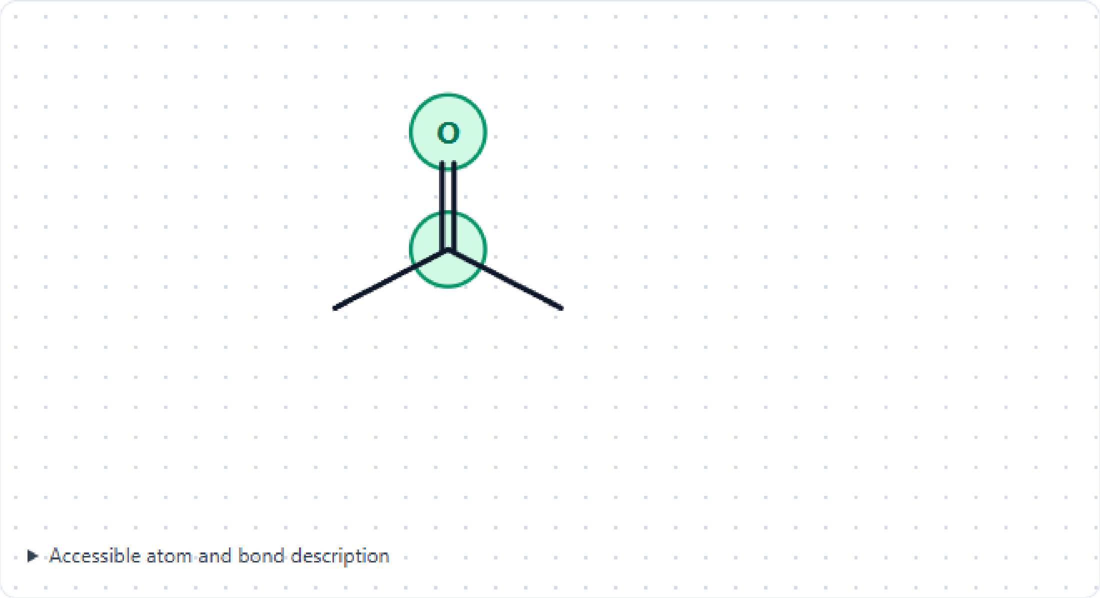
ir:step-1 | 1. Carbonyl signature

A simple saturated acyclic ketone often absorbs near 1715 cm^{-1}. The illustrative plot marks a band position, not an experimentally measured spectrum.

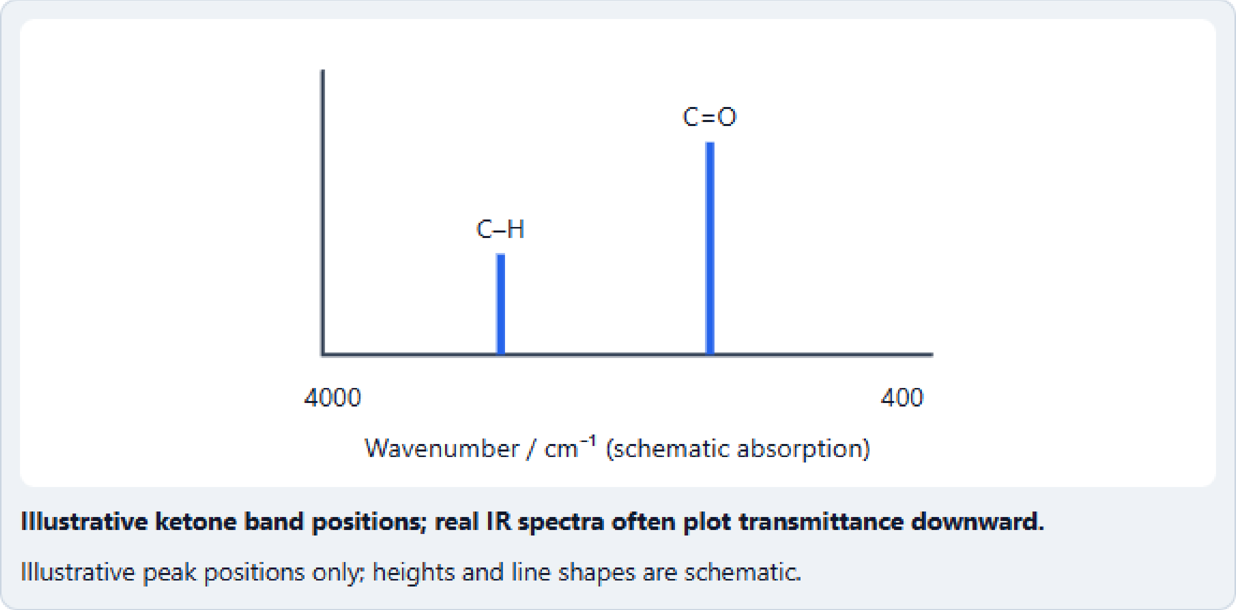
Displayed molecule ID: acetone

{"highlights": {"prin": [1, 3]}, "revealName": true}

Screenshot: ir-step-1



Screenshot: ir-step-1-diagram


C=O
C–H
4000
400
Wavenumber / cm^{-1} (schematic absorption)

Illustrative ketone band positions; real IR spectra often plot transmittance downward.

Illustrative peak positions only; heights and line shapes are schematic.

{"diagramSpecification": {"type": "spectrum", "min": 400, "max": 4000, "unit": "Wavenumber / cm^{-1} (schematic absorption)", "caption": "Illustrative ketone band positions; real IR spectra often plot transmittance downward.", "peaks": [{"x": 1715, "height": 0.95, "label": "C=O"}, {"x": 2950, "height": 0.45, "label": "C–H"}]}}

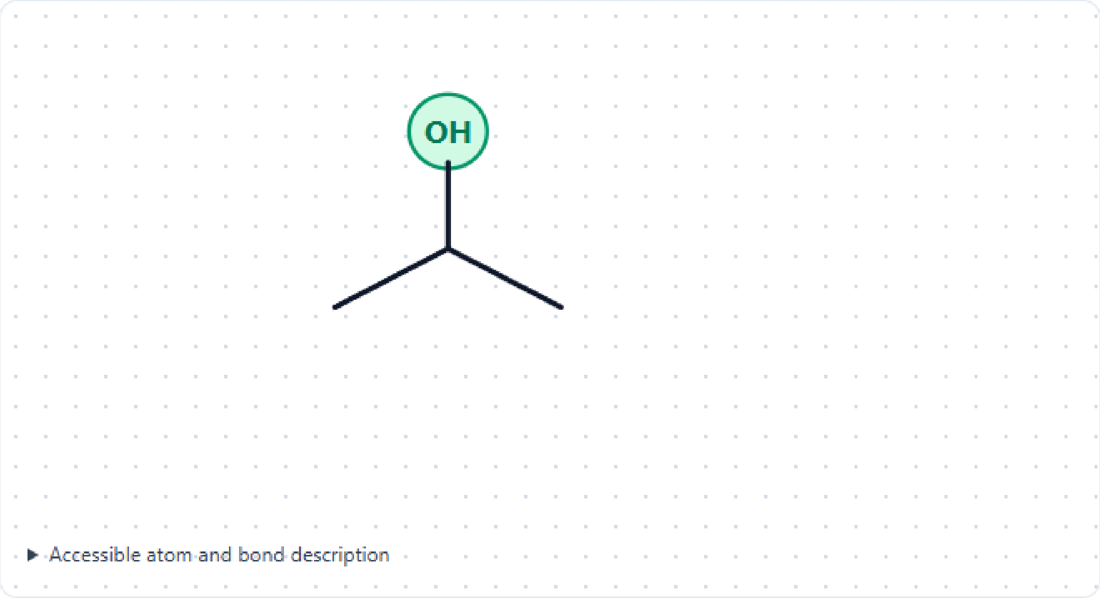
ir:step-2 | 2. Compare OH

An alcohol has an O–H vibration often broadened by hydrogen bonding. Converting acetone to propan-2-ol changes both the C=O and O–H evidence.

Displayed molecule ID: isopropanol

{"highlights": {"prin": [3]}, "revealName": true}

Screenshot: ir-step-2


OH
▶ Accessible atom and bond description

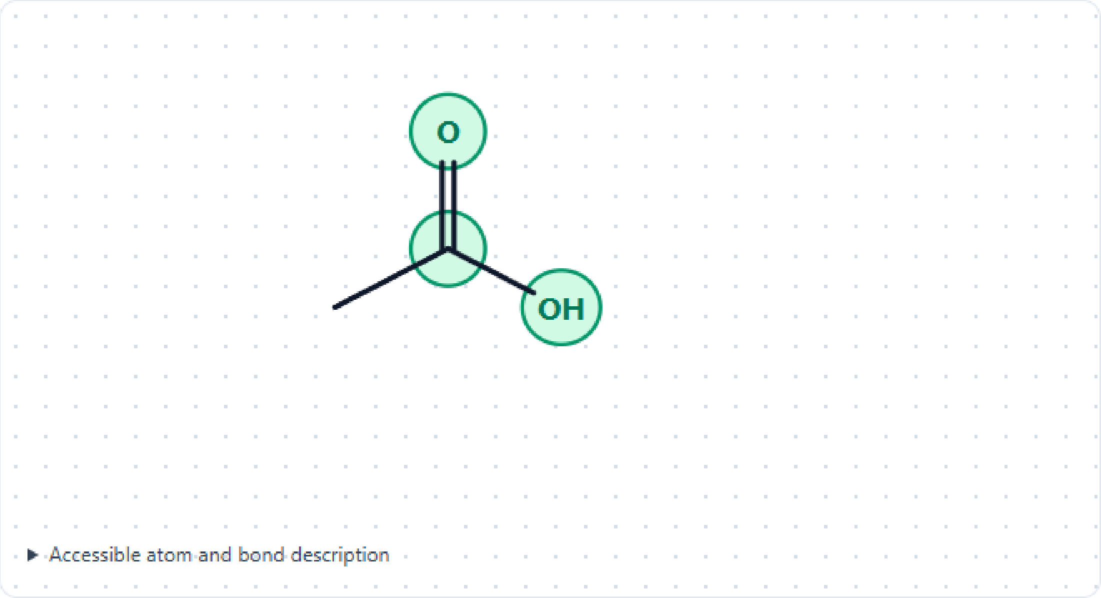
ir:step-3 | 3. Acid pattern

A carboxylic acid can show both a carbonyl band and a very broad OH band extending across roughly 2500–3300 cm^{-1}. Interpret them together.

Displayed molecule ID: aceticacid

{"highlights": {"prin": [1, 2, 3]}, "revealName": true}

Screenshot: ir-step-3


O
OH
▶ Accessible atom and bond description

uv | UV–visible: conjugation and energy

Longer wavelength means lower photon energy

uv:rule | Core Rules

UV–visible absorption promotes electrons between electronic states. In many comparable conjugated pi systems, increasing conjugation lowers the transition energy and shifts absorption to longer wavelength. Photon energy obeys $E=hc/\lambda$. Beer–Lambert law is $A=\varepsilon lc$ in its linear regime.

uv:cue | Memory Cues

More extended conjugation often gives a red shift. Doubling wavelength halves photon energy.

uv:trap | Common Mistakes / MCAT Traps

Conjugation trends compare appropriate related systems, not every possible molecule. UV absorption does not uniquely identify a structure. Beer–Lambert requires compatible units and an appropriate concentration range.

uv:app | MCAT Applications

Conjugated chromophores underlie retinal photochemistry and many biological absorbance measurements.

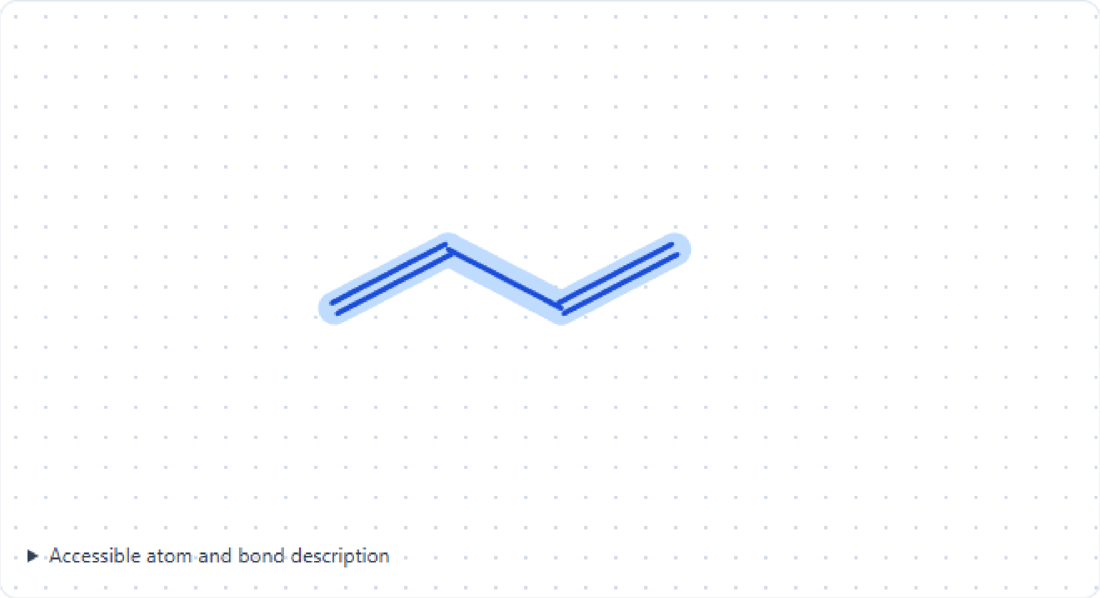
uv:step-1 | 1. Follow conjugation

Alternating C=C–C=C bonds create a connected p-orbital system. Adjacent p orbitals must remain sufficiently aligned for conjugation.

Displayed molecule ID: butadiene

{"highlights": {"path": [0, 1, 2, 3]}, "revealName": true}

Screenshot: uv-step-1



uv:step-2 | 2. Connect wavelength to energy

Relative to an isolated comparable alkene, conjugation often reduces the pi-to-pi-star transition energy. Use $E=hc/\lambda$ to translate that smaller gap into longer wavelength.

Displayed molecule ID: ethene

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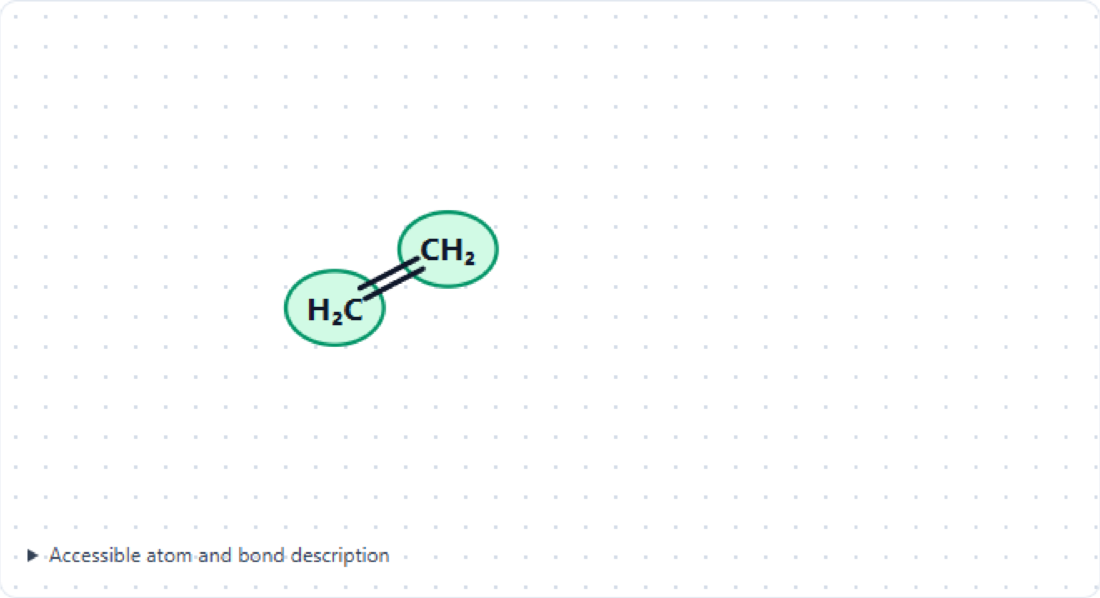
Screenshot: uv-step-2


CH₂
H₂C
▶ Accessible atom and bond description

nmr | Proton NMR: shift, integration, splitting

Three independent kinds of evidence

nmr:rule | Core Rules

Chemical shift reports environment; integration reports relative proton count; splitting reports coupling. In simple first-order cases, coupling to n equivalent neighboring protons produces n+1 lines. Equivalent protons do not split one another. Deshielded protons appear downfield at higher ppm.

nmr:cue | Memory Cues

Shift = neighborhood; area = population; splitting = neighbors.

nmr:trap | Common Mistakes / MCAT Traps

n+1 is a limited first-order rule, not universal. Nonequivalent neighbors can give more complex patterns; exchangeable OH/NH protons often do not show the expected simple coupling. Chemical equivalence depends on symmetry and timescale.

nmr:app | MCAT Applications

Typical rough proton regions: alkyl 0–3 ppm; C next to O often 3–4.5; vinylic about 4.5–6.5; aromatic 6–8.5; aldehydic 9–10; acid OH often 10–13. Use the full pattern.

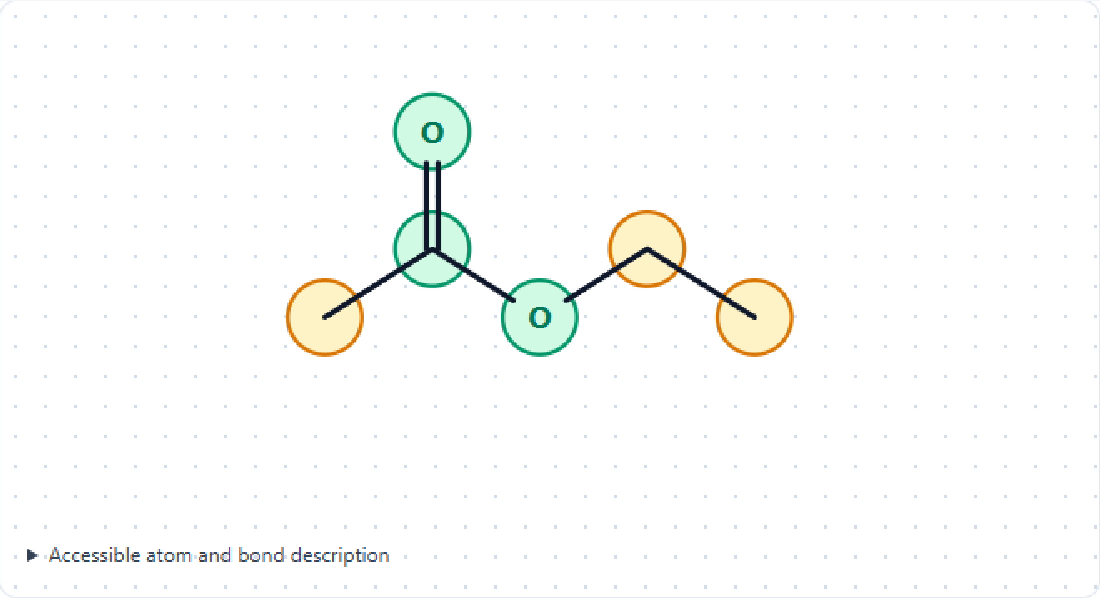
nmr:step-1 | 1. Three proton sets

Ethyl ethanoate has acyl methyl, OCH2, and terminal ethyl methyl proton sets. Their integrals are 3:2:3. The carbonyl carbon has no H.

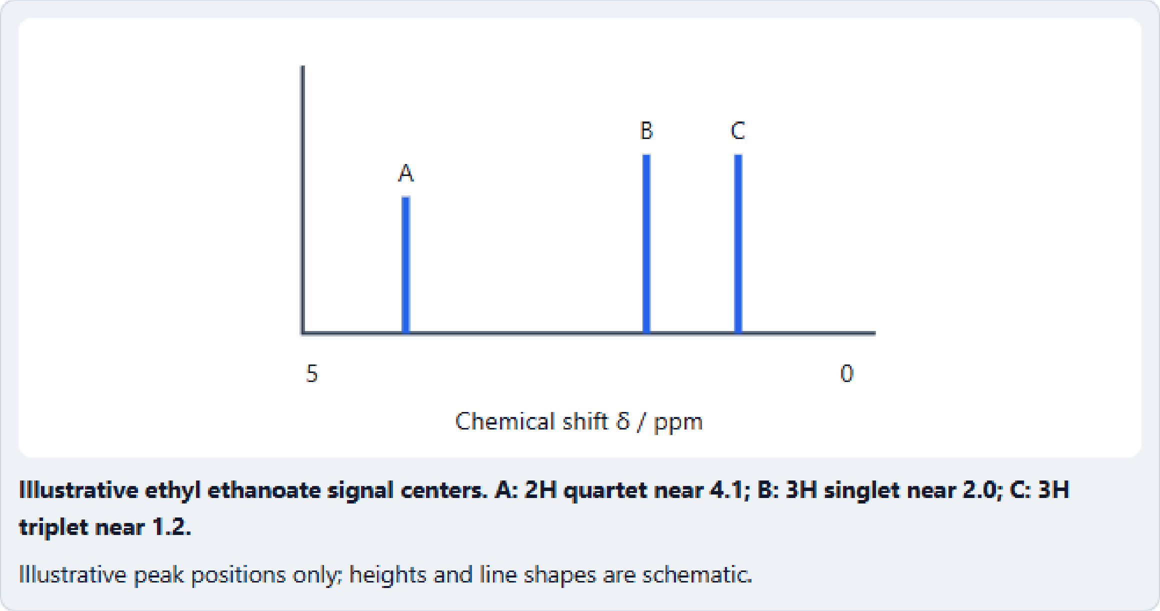
Displayed molecule ID: ethylacetate

{"highlights": {"sub": [0, 4, 5], "prin": [1, 2, 3]}, "revealName": true}

Screenshot: nmr-step-1



Screenshot: nmr-step-1-diagram



Illustrative ethyl ethanoate signal centers. A: 2H quartet near 4.1; B: 3H singlet near 2.0; C: 3H triplet near 1.2.

Illustrative peak positions only; heights and line shapes are schematic.

{"diagramSpecification": {"type": "spectrum", "min": 0, "max": 5, "unit": "Chemical shift δ / ppm", "caption": "Illustrative ethyl ethanoate signal centers. A: 2H quartet near 4.1; B: 3H singlet near 2.0; C: 3H triplet near 1.2.", "peaks": [{"x": 4.1, "height": 0.65, "label": "A"}, {"x": 2, "height": 0.85, "label": "B"}, {"x": 1.2, "height": 0.85, "label": "C"}]}}

nmr:step-2 | 2. Ethyl splitting

OCH2 couples to three neighboring methyl H atoms, giving a quartet in a simple first-order picture. The terminal CH3 couples to two methylene H atoms, giving a triplet.

Displayed molecule ID: ethylacetate

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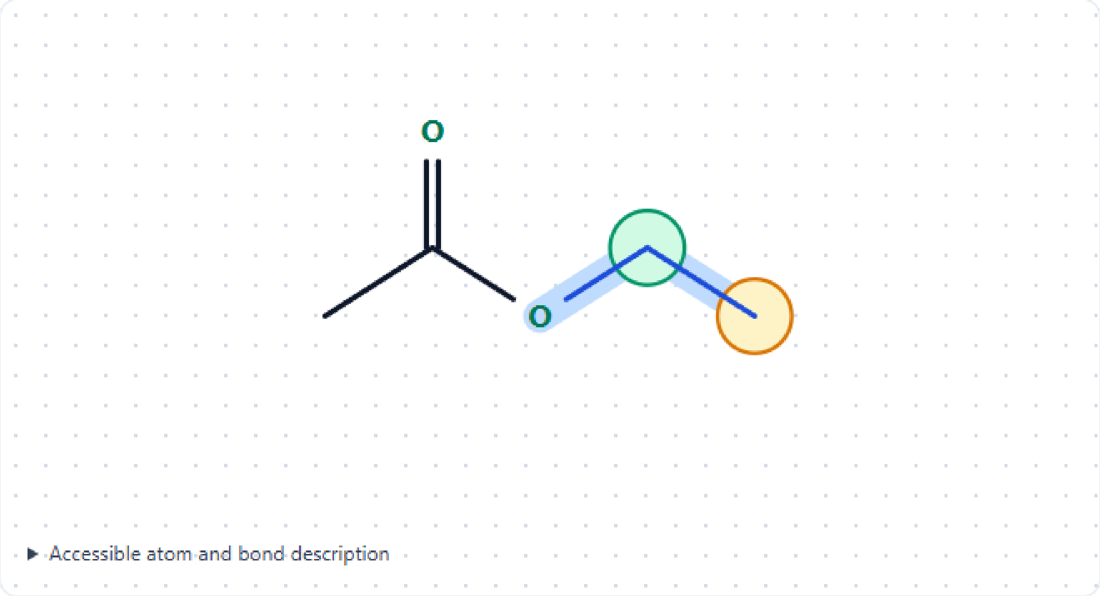
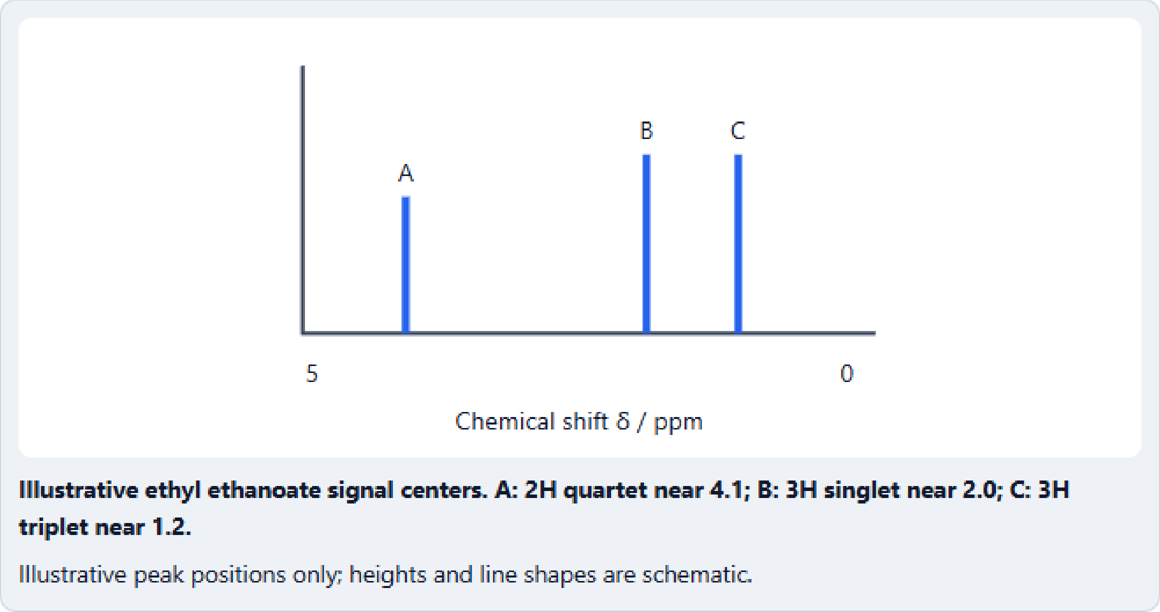
Screenshot: nmr-step-2



Screenshot: nmr-step-2-diagram



Illustrative ethyl ethanoate signal centers. A: 2H quartet near 4.1; B: 3H singlet near 2.0; C: 3H triplet near 1.2.

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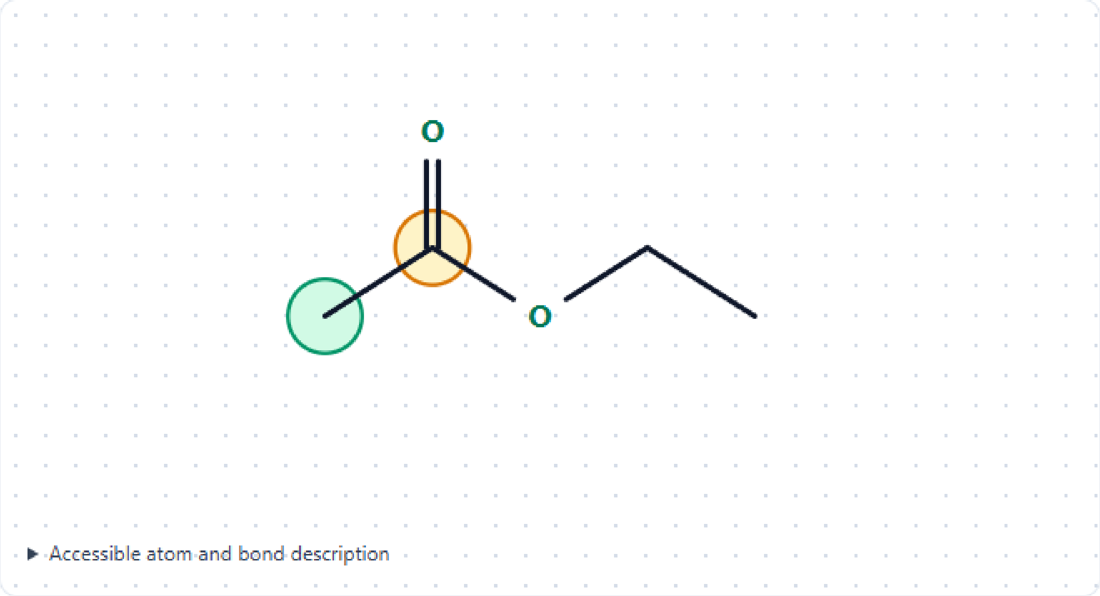
nmr:step-3 | 3. Acyl methyl

The acyl methyl has no H-bearing carbon immediately across the carbonyl bond, so it is approximately a singlet in this example. Do not apply n+1 by counting every H anywhere in the molecule.

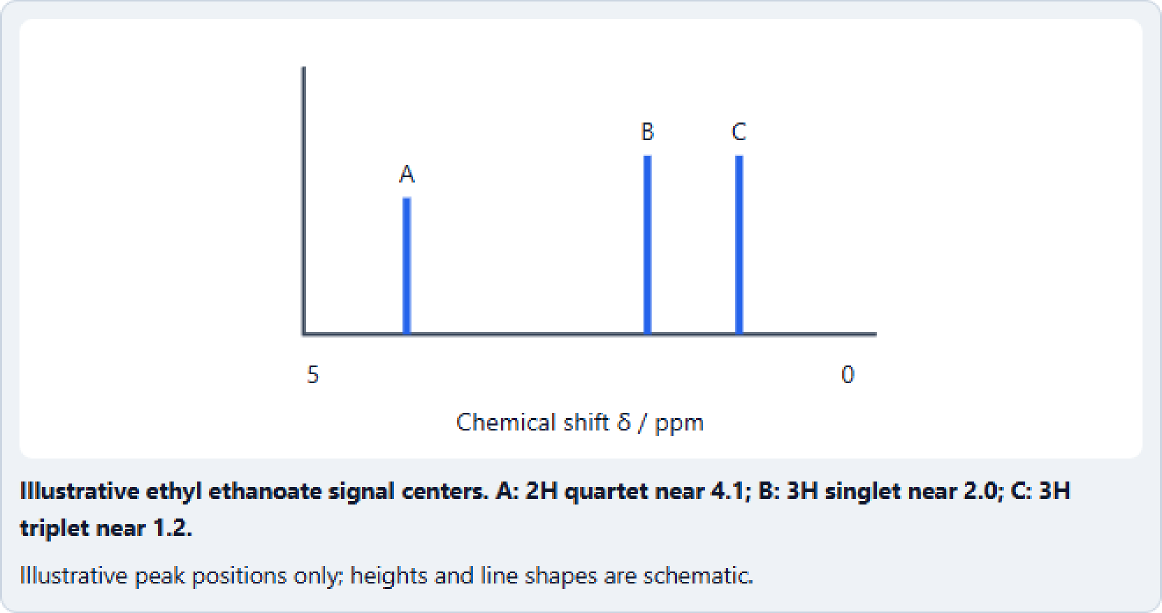
Displayed molecule ID: ethylacetate

{"highlights": {"prin": [0], "sub": [1]}, "revealName": false}

Screenshot: nmr-step-3



Screenshot: nmr-step-3-diagram



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combine | Combine formula, IR, and NMR

Independent clues should agree

combine:rule | Core Rules

For ordinary neutral closed-shell organic formulas, degree of unsaturation (DBE) = (2C+2+N−H−X)/2; O and divalent S do not enter. A ring or double bond contributes one DBE; a triple bond contributes two. Then combine IR group evidence with NMR environments and integrations.

combine:cue | Memory Cues

Formula constrains possibilities; IR finds groups; NMR organizes the skeleton.

combine:trap | Common Mistakes / MCAT Traps

A single NMR signal can represent several equivalent groups. DBE says how much unsaturation exists, not where it is. Do not force phosphorus-rich or unusual-valence formulas into this simple equation.

combine:app | MCAT Applications

Passage-based structure identification rewards eliminating structures inconsistent with even one reliable observation.

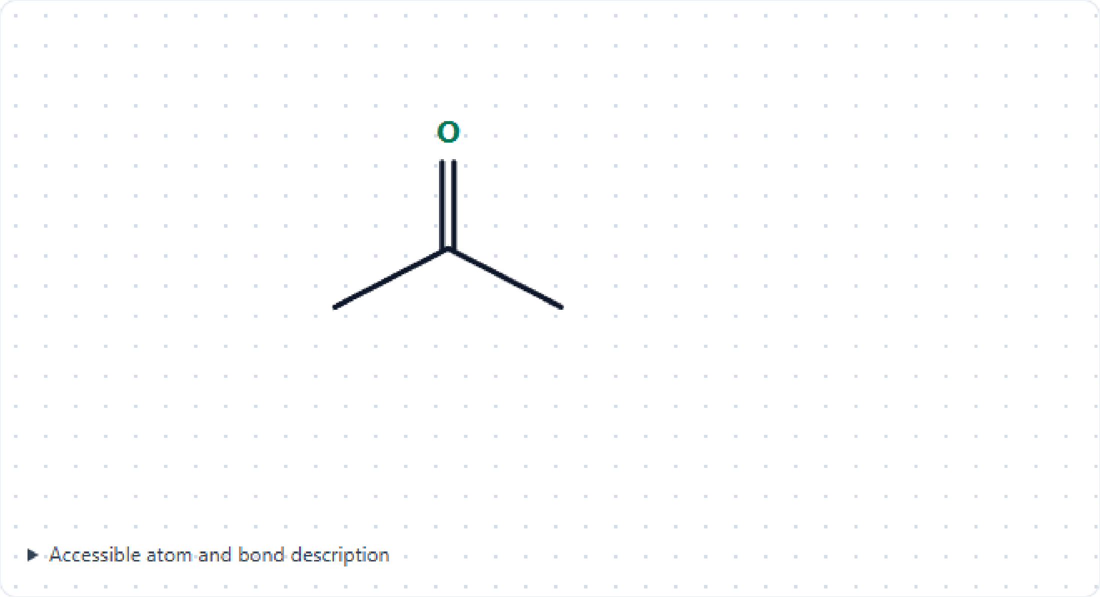
combine:step-1 | 1. Formula constraint

C_3H_6O has DBE=(2×3+2−6)/2=1. A carbonyl can account for that entire unit of unsaturation.

Displayed molecule ID: acetone

{"highlights": {}, "revealName": false}

Screenshot: combine-step-1



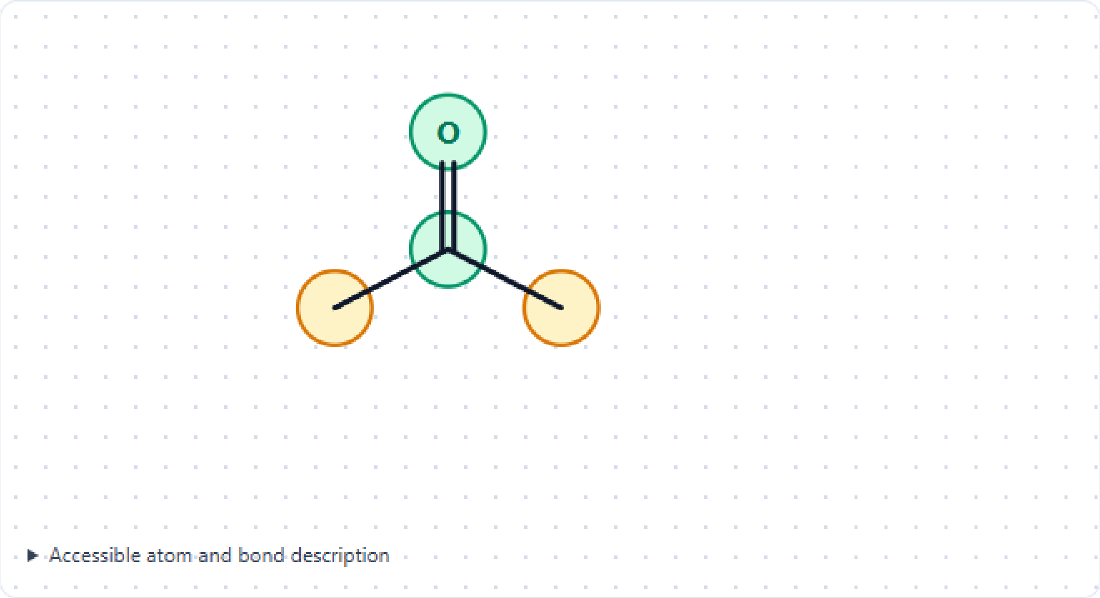
combine:step-2 | 2. Add two measurements

Suppose IR shows a ketone-like C=O band and proton NMR shows a single 6H singlet near 2 ppm. Two symmetry-equivalent methyl groups adjacent to C=O fit those data.

Displayed molecule ID: acetone

{"highlights": {"prin": [1, 3], "sub": [0, 2]}, "revealName": true}

Screenshot: combine-step-2


O
▶ Accessible atom and bond description

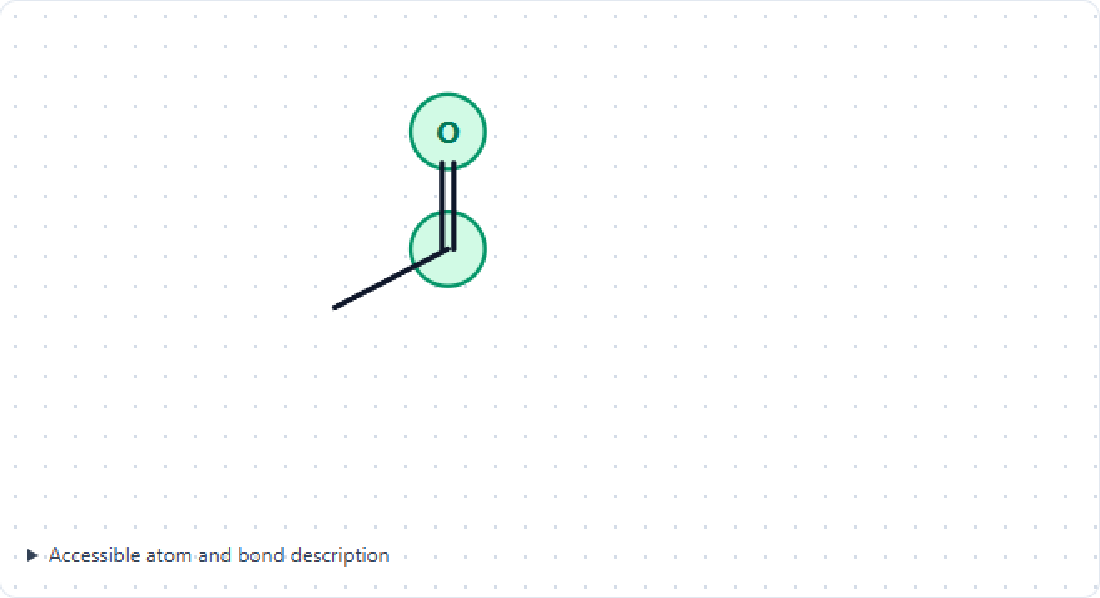
combine:step-3 | 3. Challenge an alternative

Ethanal has only two carbons and an aldehydic H; it cannot fit C3H6O or the supplied 6H singlet. Test each candidate against every observation.

Displayed molecule ID: acetaldehyde

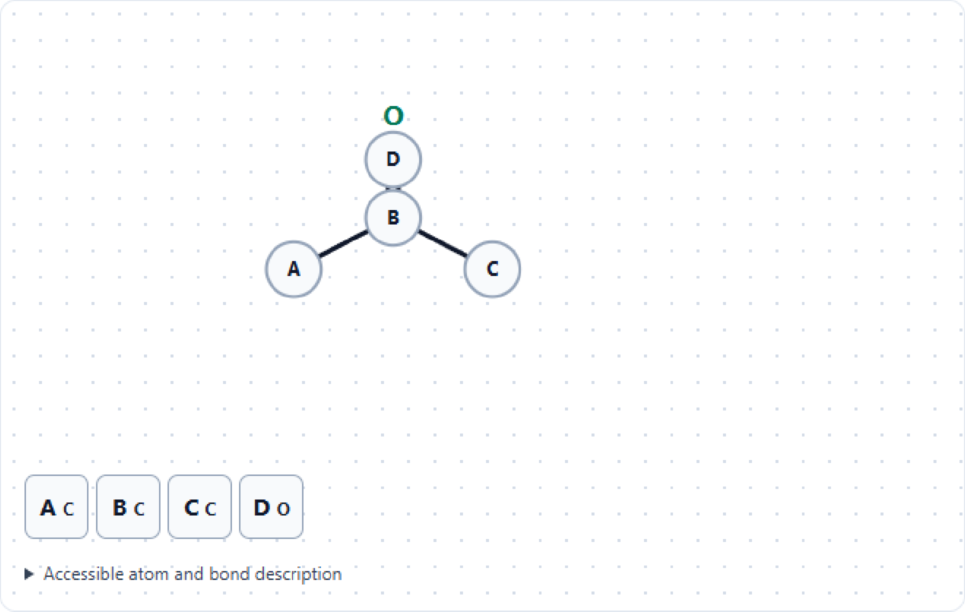
{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: combine-step-3

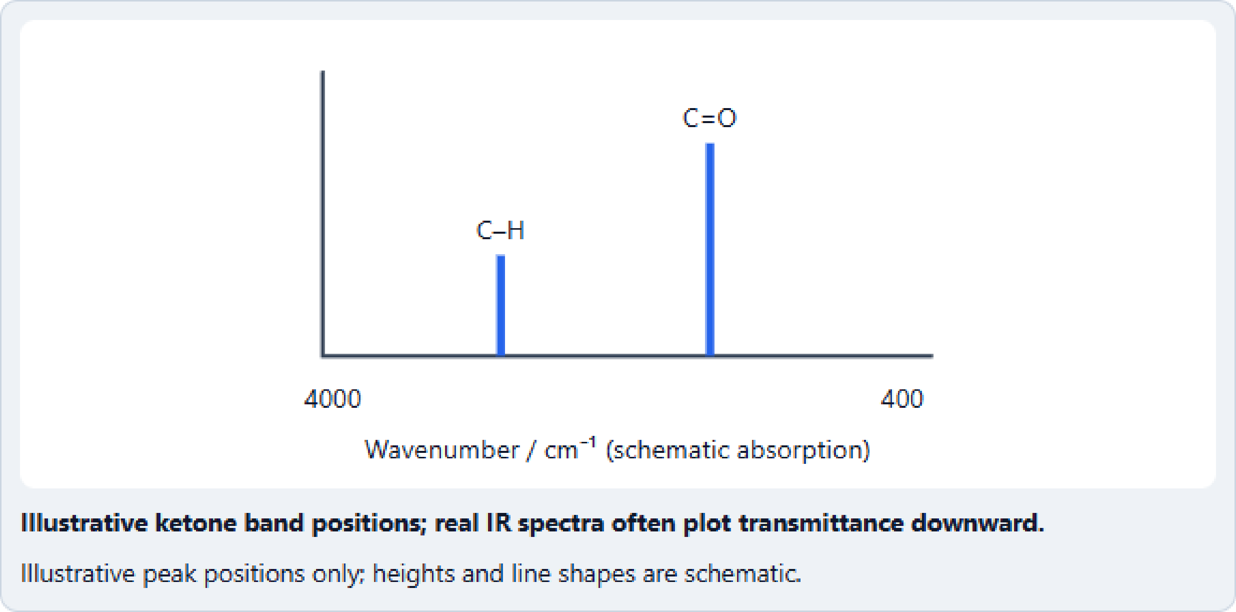

O
▶ Accessible atom and bond description

ir-read | Interpret functional-group evidence

Screenshot: ir-read-stage-1-molecule



Screenshot: ir-read-diagram



Illustrative ketone band positions; real IR spectra often plot transmittance downward.

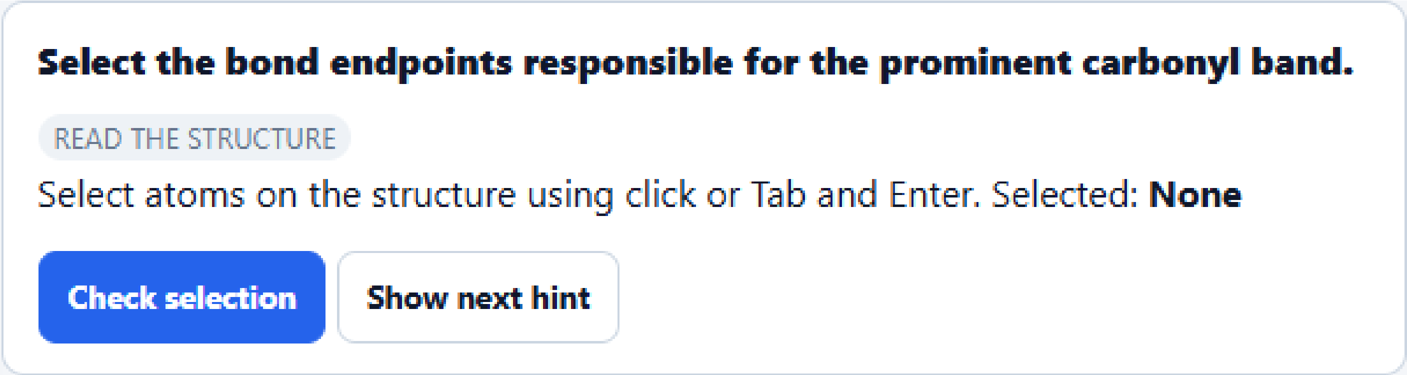
Illustrative peak positions only; heights and line shapes are schematic.

{"diagramSpecification": {"type": "spectrum", "min": 400, "max": 4000, "unit": "Wavenumber / cm^{-1} (schematic absorption)", "caption": "Illustrative ketone band positions; real IR spectra often plot transmittance downward.", "peaks": [{"x": 1715, "height": 0.95, "label": "C=O"}, {"x": 2950, " height": 0.45, "label": "C–H"}]}}

ir-read:stage-1 | Select the bond endpoints responsible for the prominent carbonyl band.

Answer type: atoms

Screenshot: ir-read-stage-1-question


Select the bond endpoints responsible for the prominent carbonyl band.

READ THE STRUCTURE

Select atoms on the structure using click or Tab and Enter. Selected: **None**

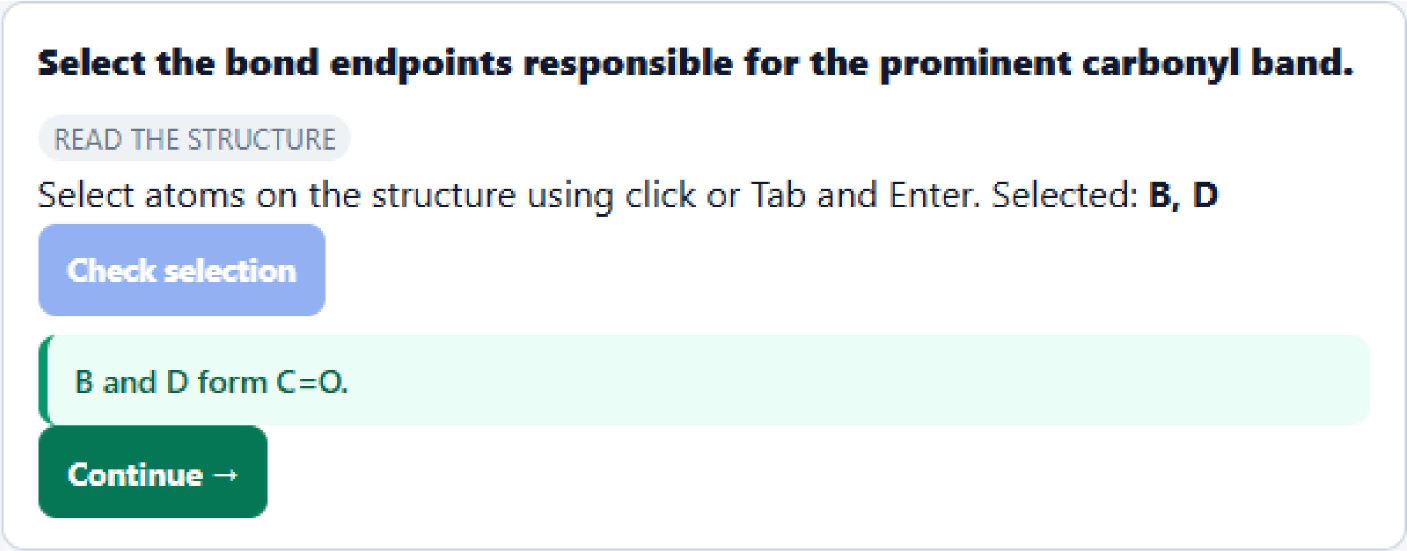
{"authorAnswer": [1, 3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 3]}}

Hint 1: Find the double bond to O.

Hint 2: Select both carbon and oxygen.

Explanation: B and D form C=O.

Screenshot: ir-read-stage-1-feedback


Select the bond endpoints responsible for the prominent carbonyl band.

READ THE STRUCTURE

Select atoms on the structure using click or Tab and Enter. Selected: **B, D**

B and D form C=O.

ir-read:stage-2 | An IR-active vibration must change...

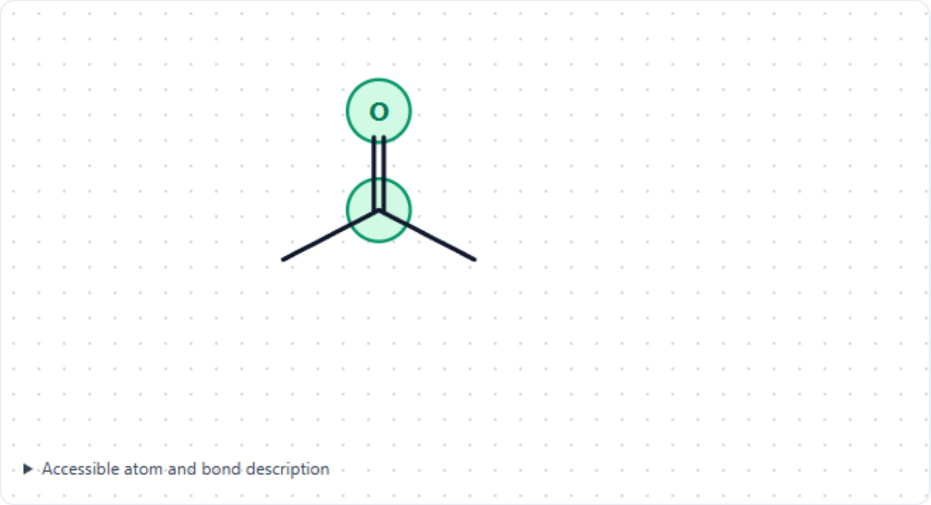
Answer type: choice

Option 0 (A): Dipole moment

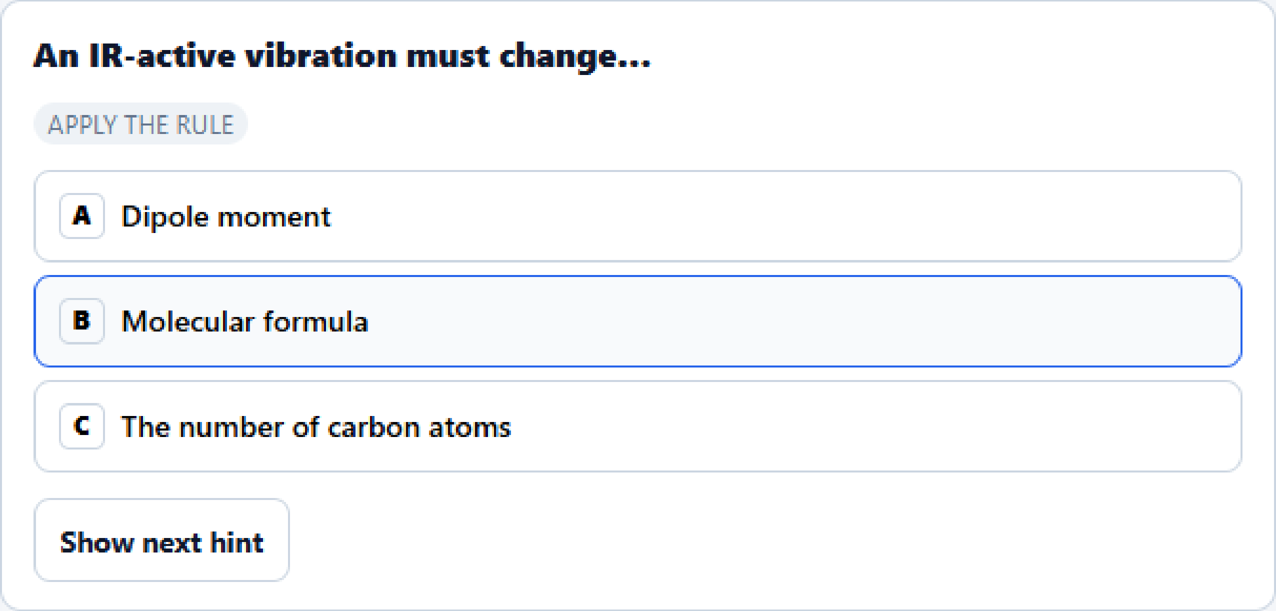
Option 1 (B): Molecular formula

Option 2 (C): The number of carbon atoms

Screenshot: ir-read-stage-2-molecule


O
▶ Accessible atom and bond description

Screenshot: ir-read-stage-2-question



An IR-active vibration must change...

APPLY THE RULE

A
B
C
Show next hint

A Dipole moment

B Molecular formula

C The number of carbon atoms

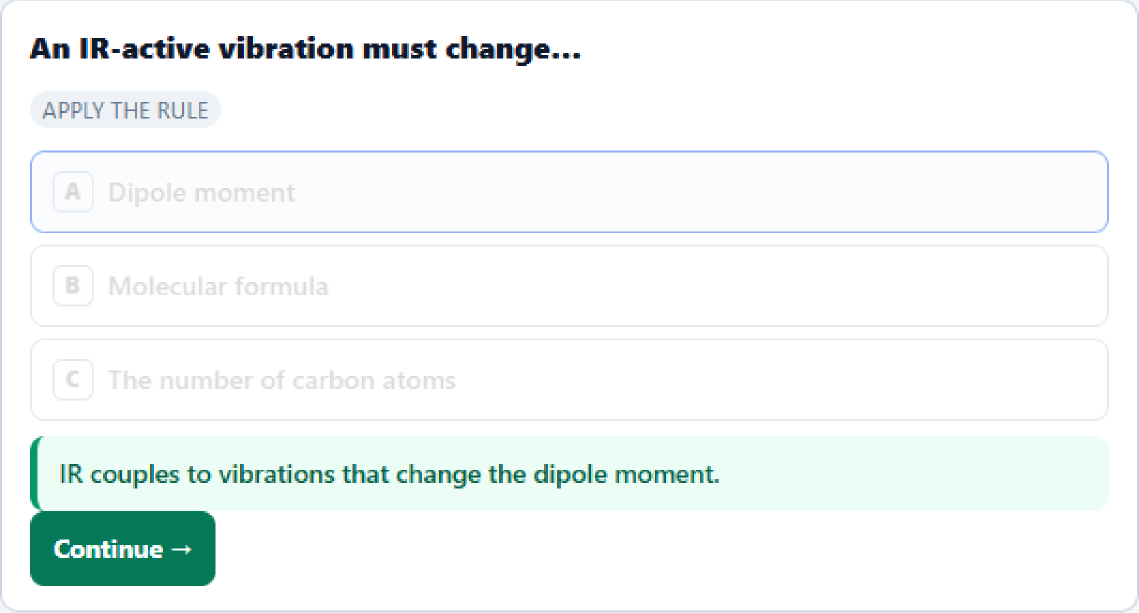
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Think about interaction with oscillating electric fields.

Hint 2: Not every possible vibration absorbs IR.

Explanation: IR couples to vibrations that change the dipole moment.

Screenshot: ir-read-stage-2-feedback



An IR-active vibration must change...

APPLY THE RULE

A Dipole moment

B Molecular formula

C The number of carbon atoms

IR couples to vibrations that change the dipole moment.

ir-read:stage-3 | A broad 3200–3600 cm^{-1} band appearing after reduction supports...

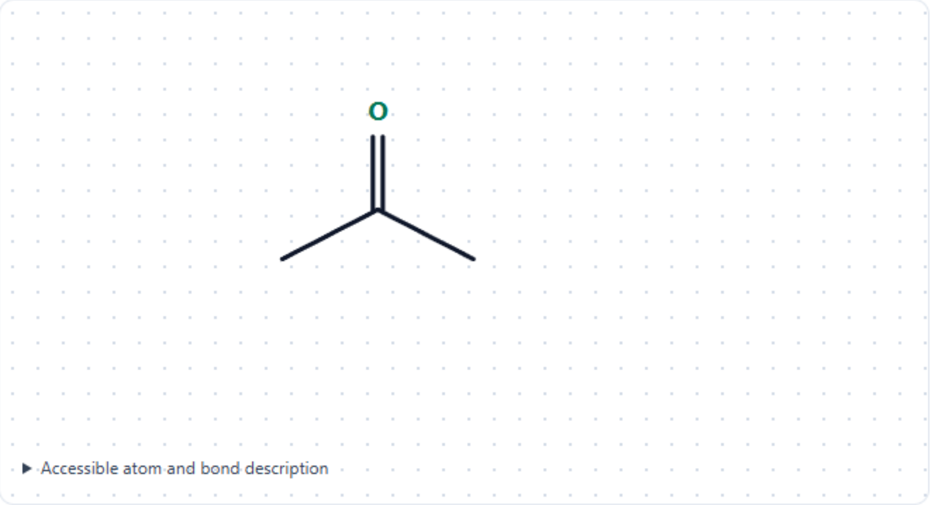
Answer type: choice

Option 0 (A): An alcohol OH

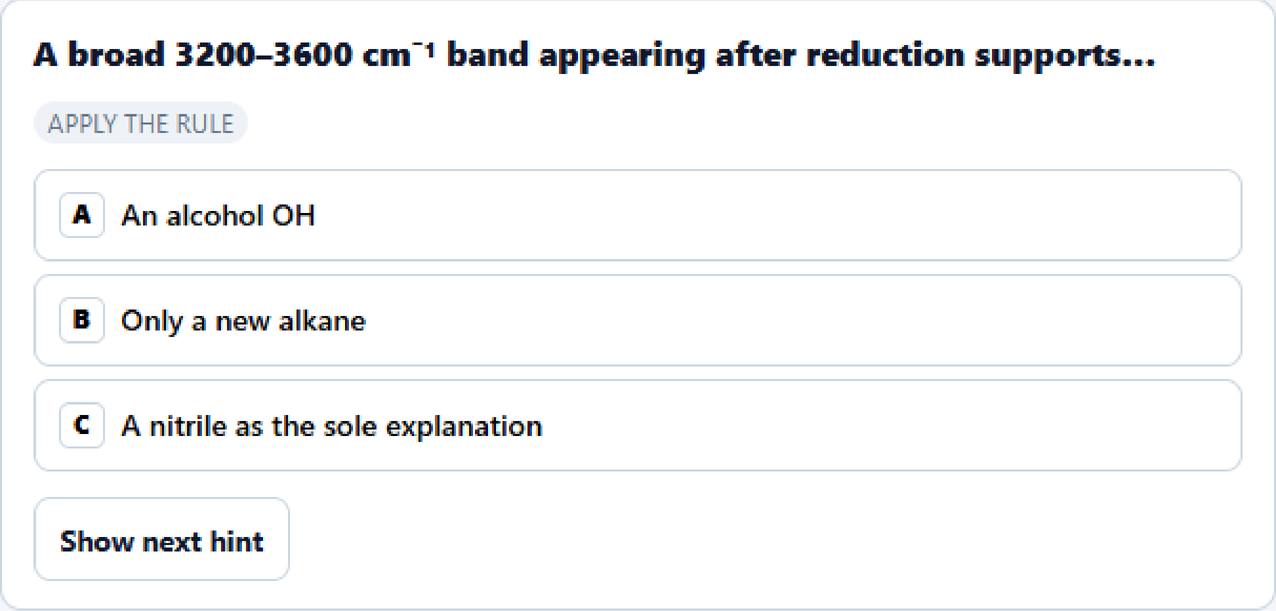
Option 1 (B): Only a new alkane

Option 2 (C): A nitrile as the sole explanation

Screenshot: ir-read-stage-3-molecule



Screenshot: ir-read-stage-3-question



A broad 3200–3600 cm^{-1} band appearing after reduction supports...

A An alcohol OH

B Only a new alkane

C A nitrile as the sole explanation

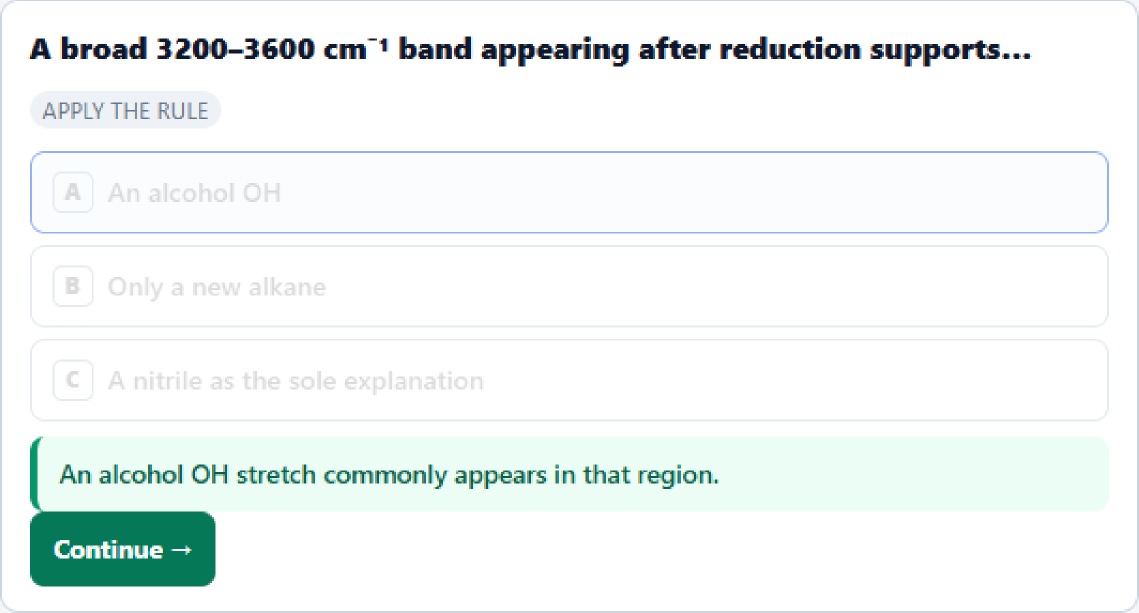
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Reduction retains oxygen.

Hint 2: C=O becomes C–OH.

Explanation: An alcohol OH stretch commonly appears in that region.

Screenshot: ir-read-stage-3-feedback



A broad 3200–3600 cm^{-1} band appearing after reduction supports...

A An alcohol OH

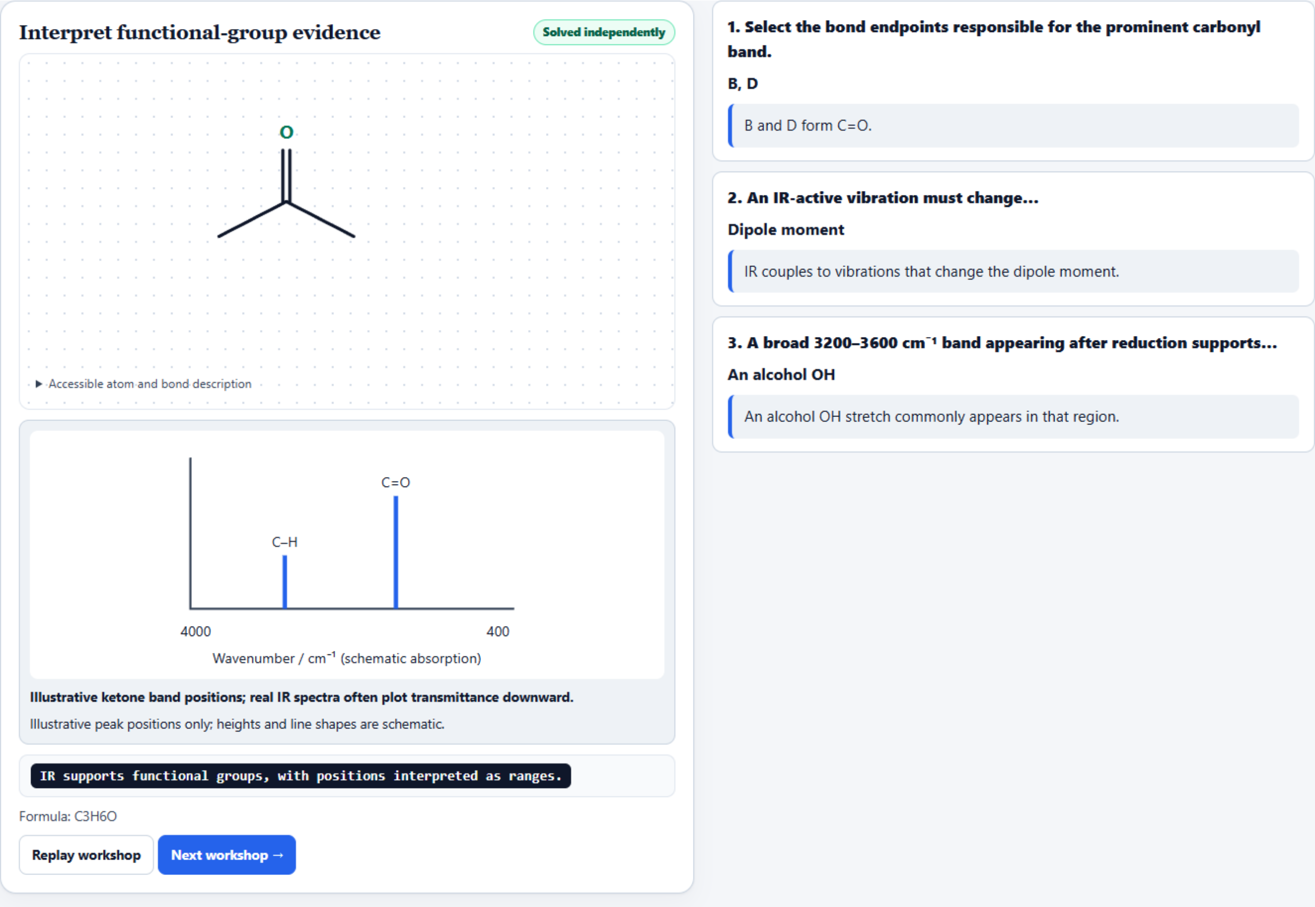
B Only a new alkane

C A nitrile as the sole explanation

An alcohol OH stretch commonly appears in that region.

Completion summary: IR supports functional groups, with positions interpreted as ranges.

Screenshot: ir-read-complete


O
C–H
C=O
4000
400
Wavenumber / cm^{-1} (schematic absorption)
▶ Accessible atom and bond description
Solved independently
Replay workshop
Next workshop →

Interpret functional-group evidence

Illustrative ketone band positions; real IR spectra often plot transmittance downward.

Illustrative peak positions only; heights and line shapes are schematic.

`IR supports functional groups, with positions interpreted as ranges.`

Formula: C3H6O

1. Select the bond endpoints responsible for the prominent carbonyl band.

B, D

> B and D form C=O.

2. An IR-active vibration must change…

Dipole moment

> IR couples to vibrations that change the dipole moment.

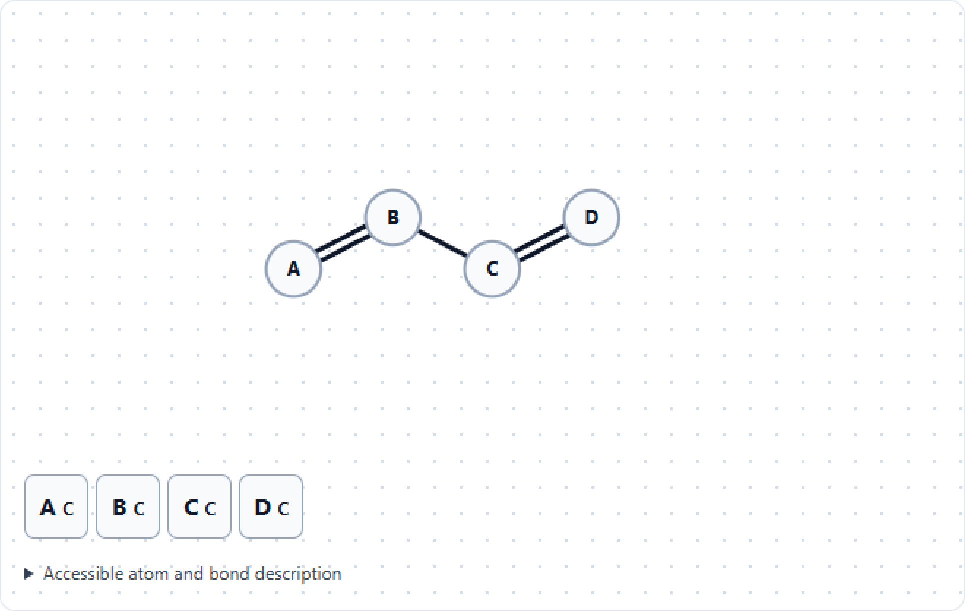
3. A broad 3200–3600 cm^{-1} band appearing after reduction supports…

An alcohol OH

> An alcohol OH stretch commonly appears in that region.

uv-energy | Connect conjugation and absorbance

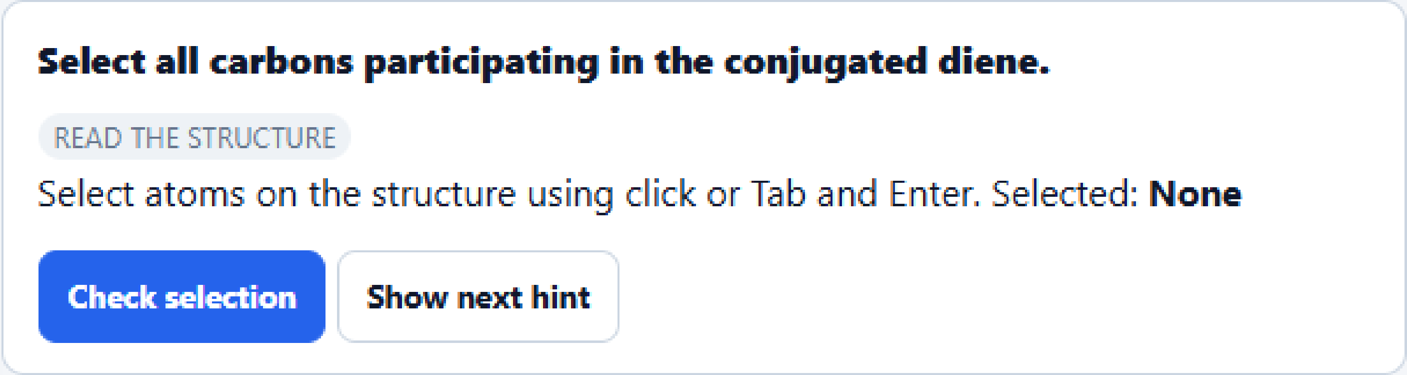
Screenshot: uv-energy-stage-1-molecule


B
D
A
C
A C
B C
C C
D C
▸ Accessible atom and bond description

uv-energy:stage-1 | Select all carbons participating in the conjugated diene.

Answer type: atoms

Screenshot: uv-energy-stage-1-question



Select all carbons participating in the conjugated diene.

READ THE STRUCTURE

Select atoms on the structure using click or Tab and Enter. Selected: **None**

Check selection
Show next hint

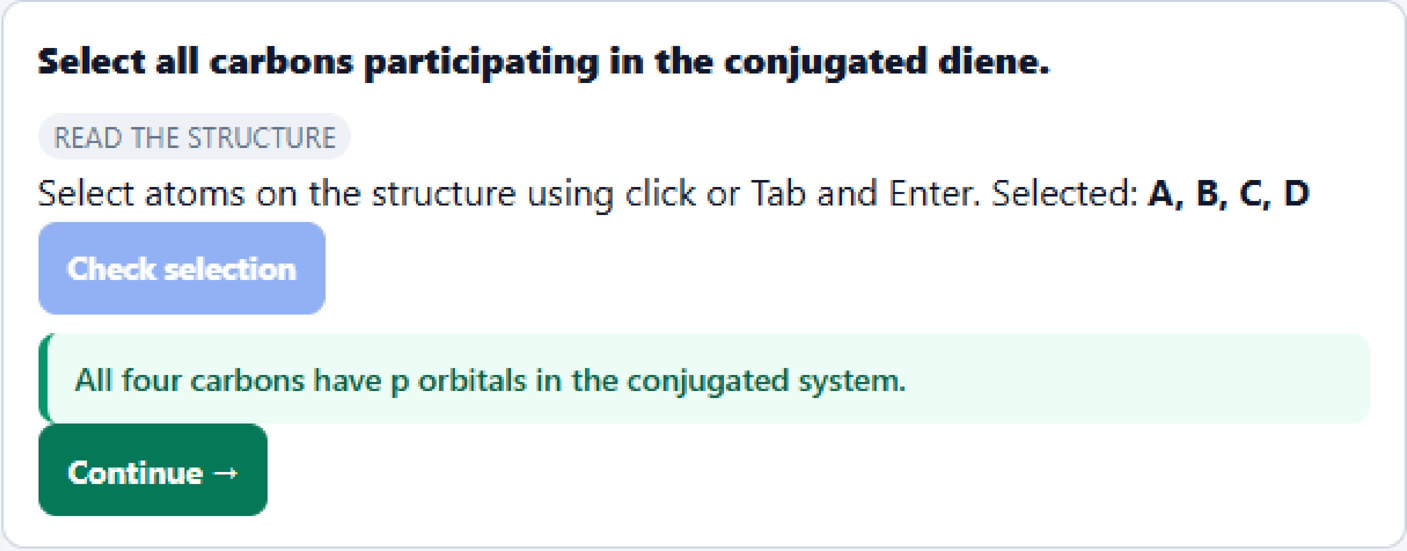
{"authorAnswer": [0, 1, 2, 3], "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3]}}

Hint 1: Follow the alternating double–single–double pattern.

Hint 2: Include the carbons of both double bonds.

Explanation: All four carbons have p orbitals in the conjugated system.

Screenshot: uv-energy-stage-1-feedback



Select all carbons participating in the conjugated diene.

READ THE STRUCTURE

Select atoms on the structure using click or Tab and Enter. Selected: **A, B, C, D**

Check selection

All four carbons have p orbitals in the conjugated system.

Continue →

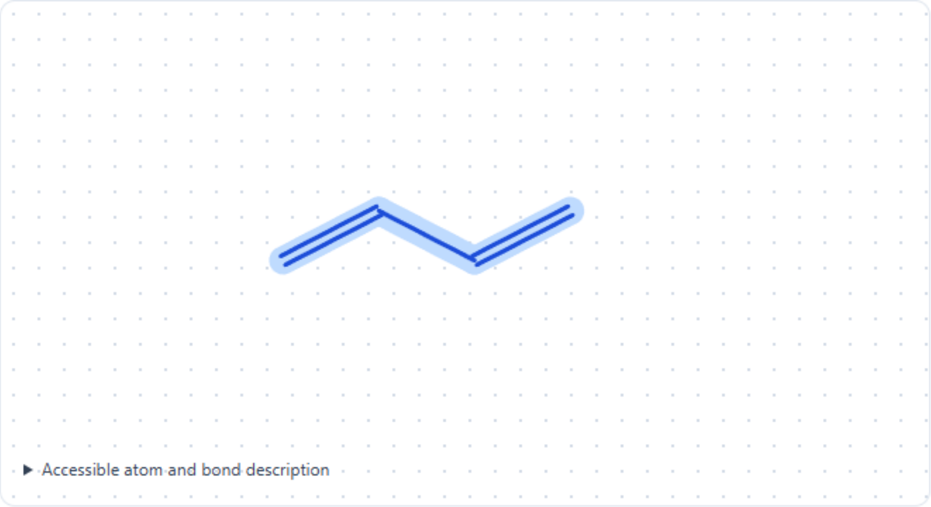
uv-energy:stage-2 | For comparable pi systems, increased conjugation often shifts absorption to...

Answer type: choice

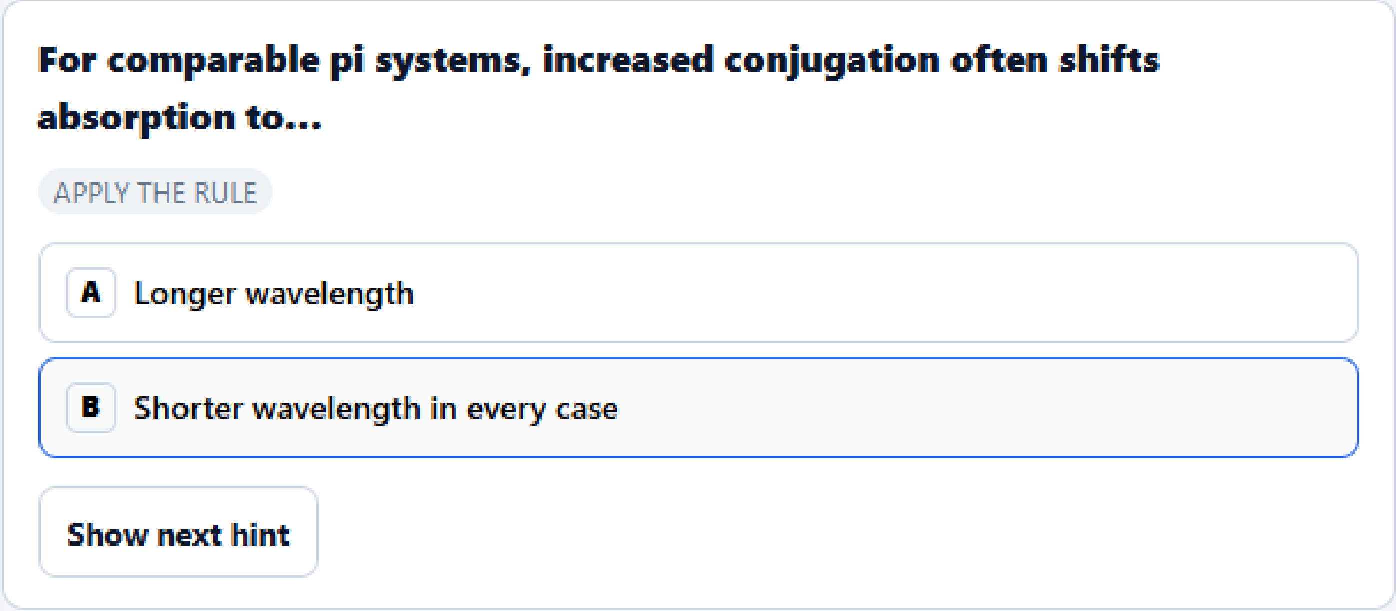
Option 0 (A): Longer wavelength

Option 1 (B): Shorter wavelength in every case

Screenshot: uv-energy-stage-2-molecule


▶ Accessible atom and bond description

Screenshot: uv-energy-stage-2-question



For comparable pi systems, increased conjugation often shifts absorption to...

APPLY THE RULE

A Longer wavelength

B Shorter wavelength in every case

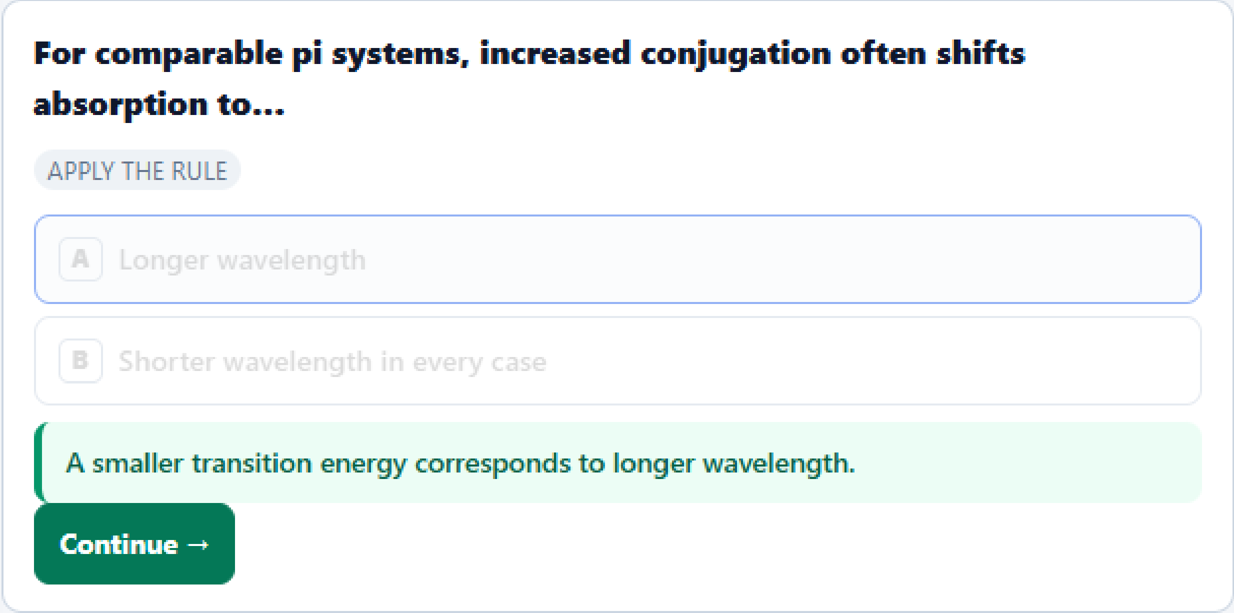
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use E=hc/λ.

Hint 2: Lower energy and longer wavelength are inversely related.

Explanation: A smaller transition energy corresponds to longer wavelength.

Screenshot: uv-energy-stage-2-feedback



For comparable pi systems, increased conjugation often shifts absorption to...

APPLY THE RULE

A Longer wavelength

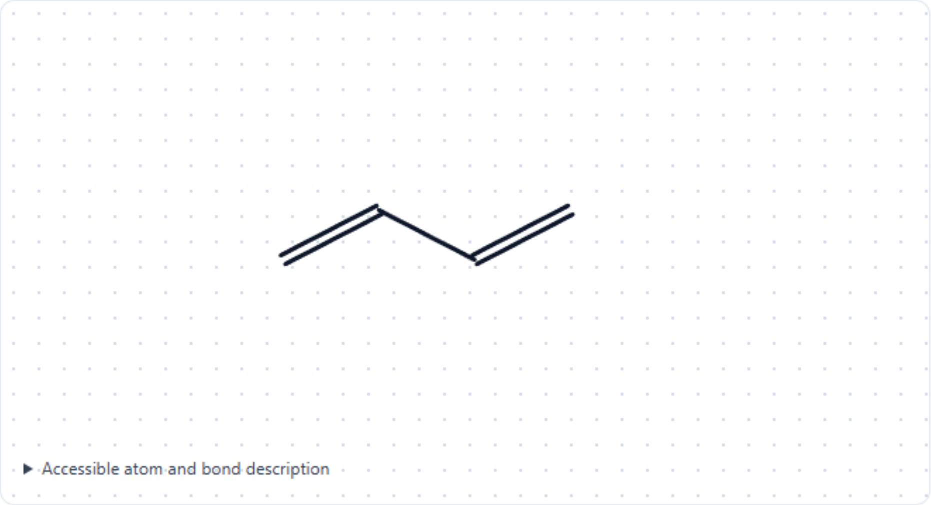
B Shorter wavelength in every case

A smaller transition energy corresponds to longer wavelength.

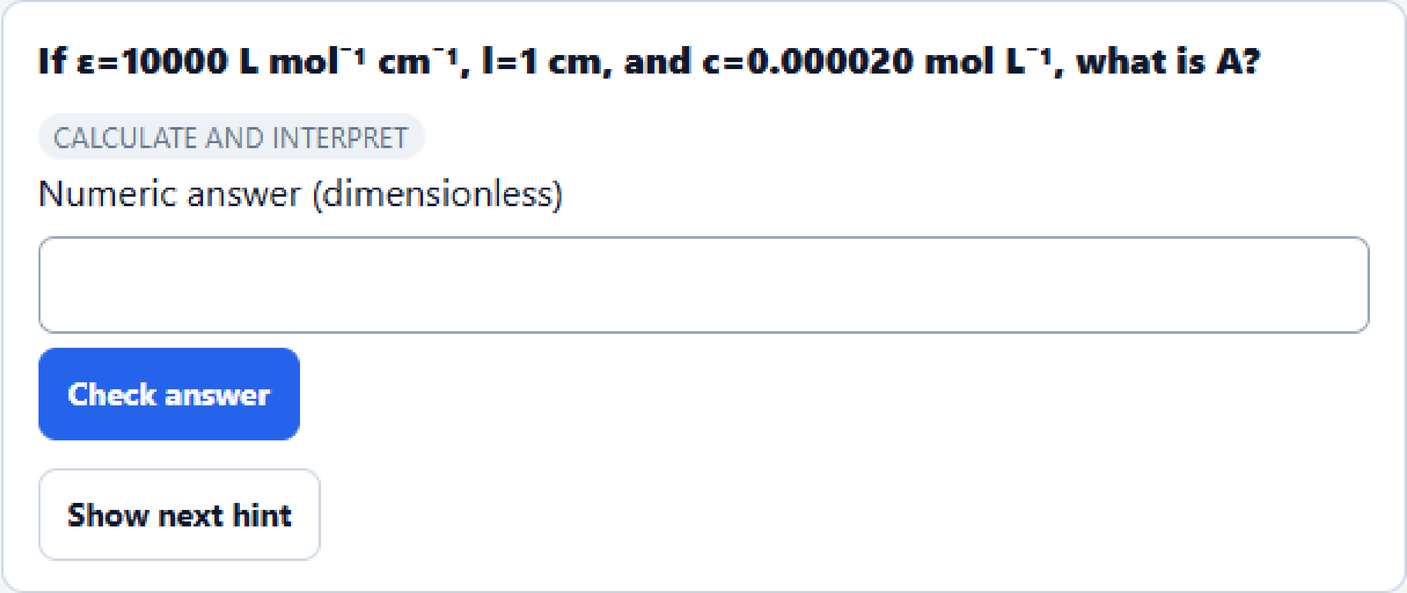
uv-energy:stage-3 | If ε=10000 L mol^{-1} cm^{-1}, l=1 cm, and c=0.000020 mol L^{-1}, what is A?

Answer type: number | unit: dimensionless

Screenshot: uv-energy-stage-3-molecule


▶ Accessible atom and bond description

Screenshot: uv-energy-stage-3-question



If ε=10000 L mol^{-1} cm^{-1}, l=1 cm, and c=0.000020 mol L^{-1}, what is A?

CALCULATE AND INTERPRET

Numeric answer (dimensionless)

Check answer
Show next hint

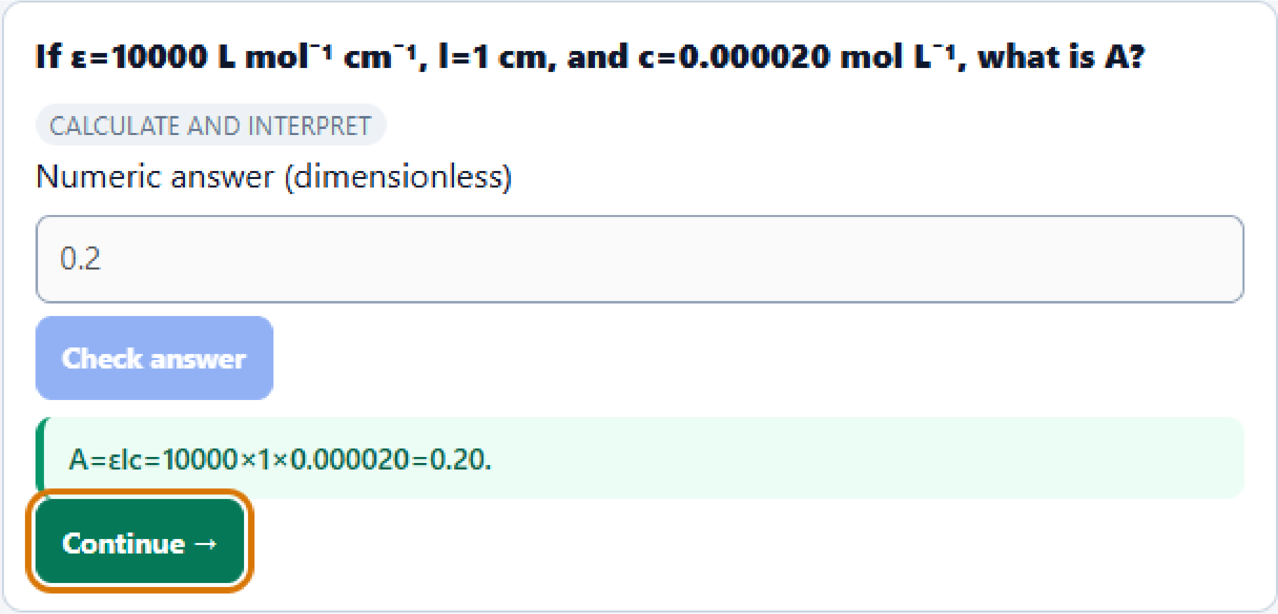
{"authorAnswer": 0.2, "tolerance": 0.001, "highlightsAfterCorrectAnswer": {}}

Hint 1: Multiply the three quantities.

Hint 2: The specified units cancel.

Explanation: $A = \varepsilon lc = 10000 \times 1 \times 0.000020 = 0.20$.

Screenshot: uv-energy-stage-3-feedback



If ε=10000 L mol^{-1} cm^{-1}, l=1 cm, and c=0.000020 mol L^{-1}, what is A?

CALCULATE AND INTERPRET

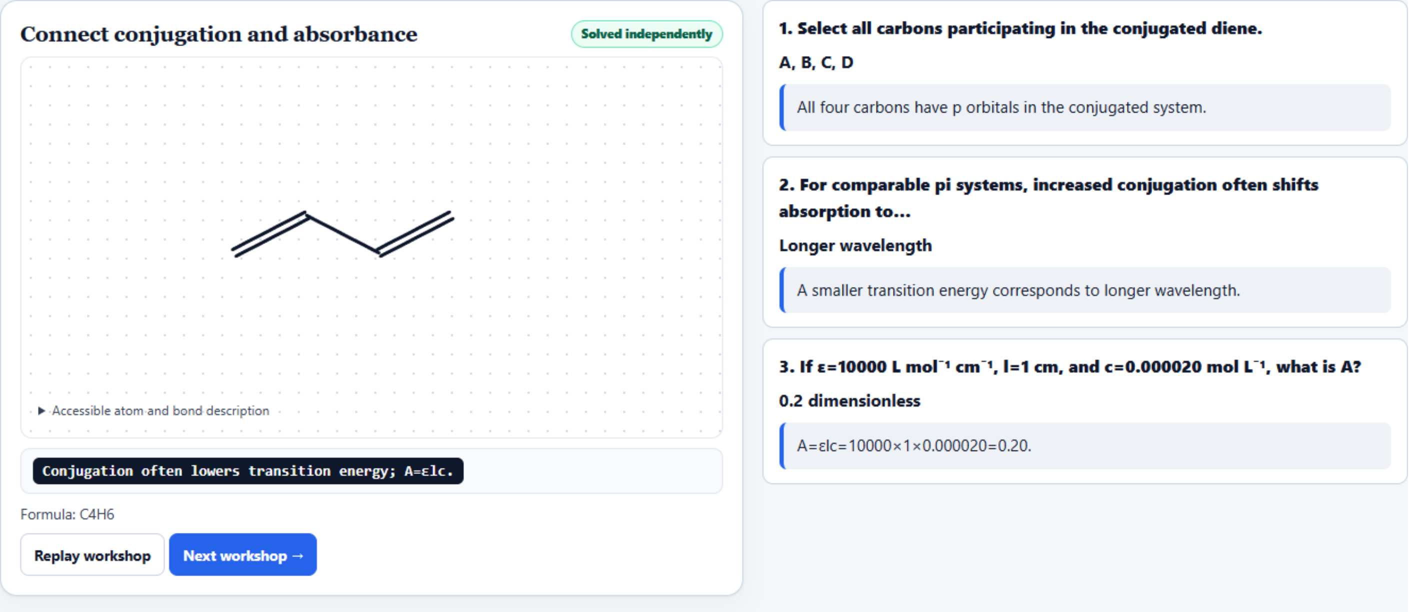
Numeric answer (dimensionless)

0.2

A=εlc=10000×1×0.000020=0.20.

Completion summary: Conjugation often lowers transition energy; A=εlc.

Screenshot: uv-energy-complete



Connect conjugation and absorbance

Solved independently

▸ Accessible atom and bond description

`Conjugation often lowers transition energy; A=εlc.`

Formula: C_4H_6

1. Select all carbons participating in the conjugated diene.

A, B, C, D

All four carbons have p orbitals in the conjugated system.

2. For comparable pi systems, increased conjugation often shifts absorption to...

Longer wavelength

A smaller transition energy corresponds to longer wavelength.

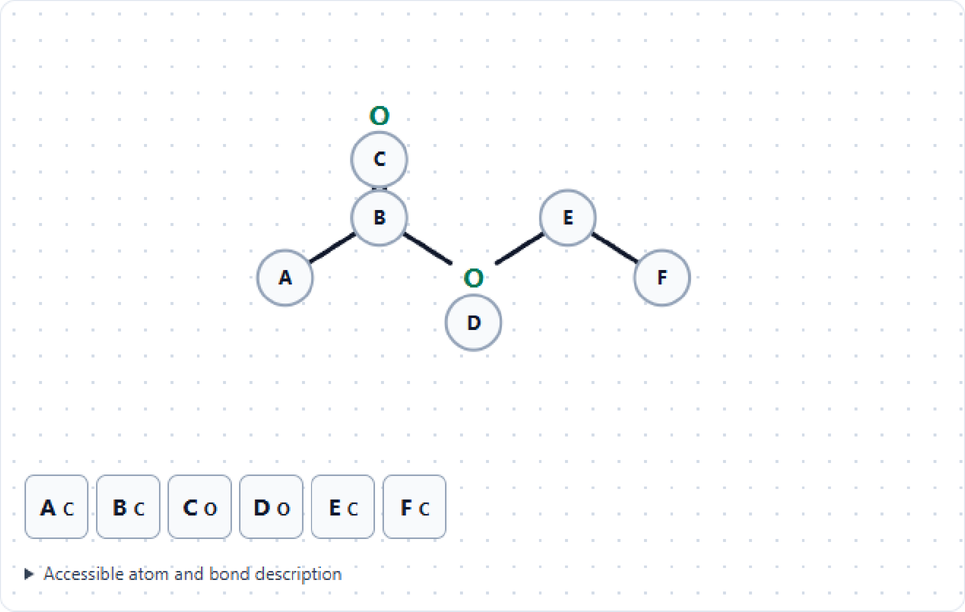
3. If ε=10000 L mol^{-1} cm^{-1}, l=1 cm, and c=0.000020 mol L^{-1}, what is A?

0.2 dimensionless

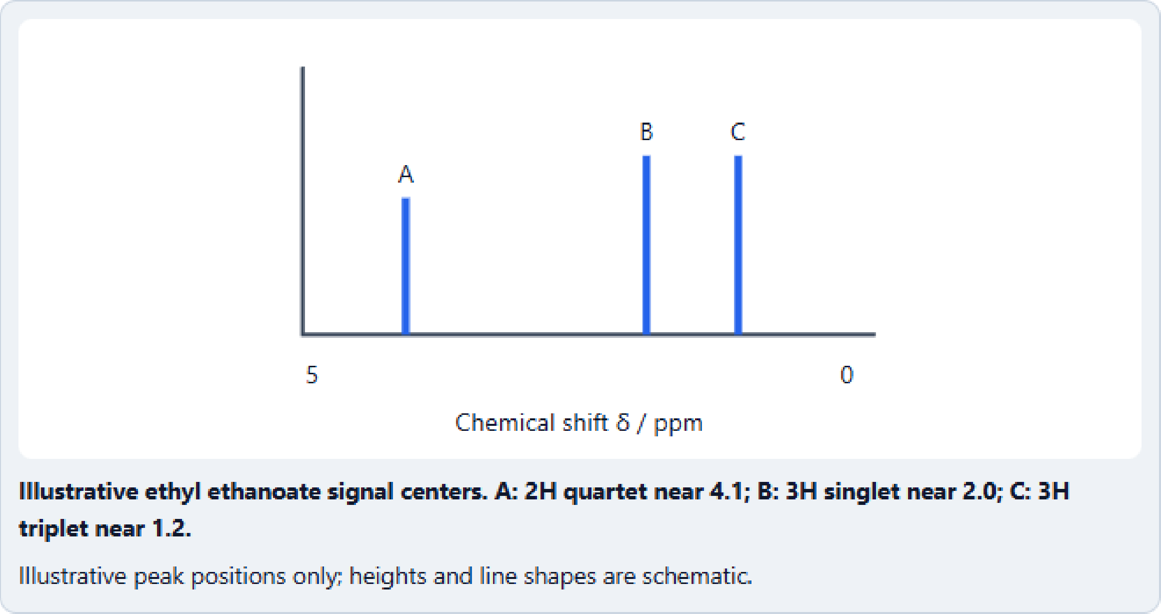
A=εlc=10000×1×0.000020=0.20.

nmr-ethyl | Assign an ethyl-group pattern

Screenshot: nmr-ethyl-stage-1-molecule


O
C
B
E
A
O
F
D
A C
B C
C O
D O
E C
F C
▸ Accessible atom and bond description

Screenshot: nmr-ethyl-diagram


B
C
A
5
0
Chemical shift δ / ppm

Illustrative ethyl ethanoate signal centers. A: 2H quartet near 4.1; B: 3H singlet near 2.0; C: 3H triplet near 1.2.

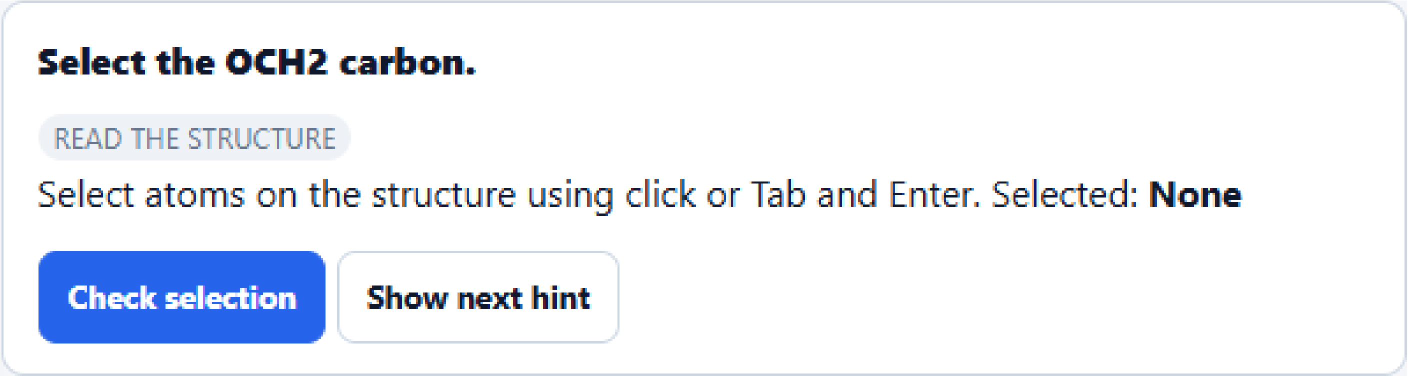
Illustrative peak positions only; heights and line shapes are schematic.

{"diagramSpecification": {"type": "spectrum", "min": 0, "max": 5, "unit": "Chemical shift δ / ppm", "caption": "Illustrative ethyl ethanoate signal centers. A: 2H quartet near 4.1; B: 3H singlet near 2.0; C: 3H triplet near 1.2.", "peaks": [{"x": 4.1, "height": 0.65, "label": "A"}, {"x": 2, "height": 0.85, "label": "B"}, {"x": 1.2, "height": 0.85, "label": "C"}]}}

nmr-ethyl:stage-1 | Select the OCH2 carbon.

Answer type: atoms

Screenshot: nmr-ethyl-stage-1-question


Select the OCH2 carbon.

READ THE STRUCTURE

Select atoms on the structure using click or Tab and Enter. Selected: **None**

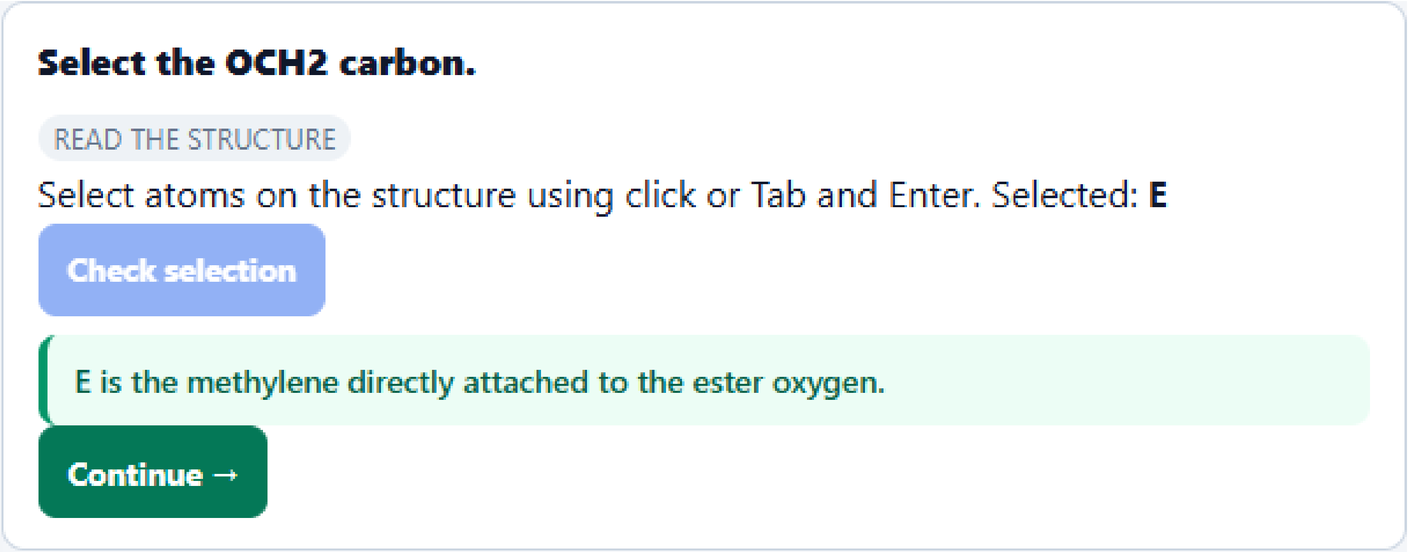
{"authorAnswer": [4], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [4], "sub": [5]}}

Hint 1: Find O–CH2–CH3.

Hint 2: Choose the carbon nearest O on the ethyl side.

Explanation: E is the methylene directly attached to the ester oxygen.

Screenshot: nmr-ethyl-stage-1-feedback


Select the OCH2 carbon.

READ THE STRUCTURE

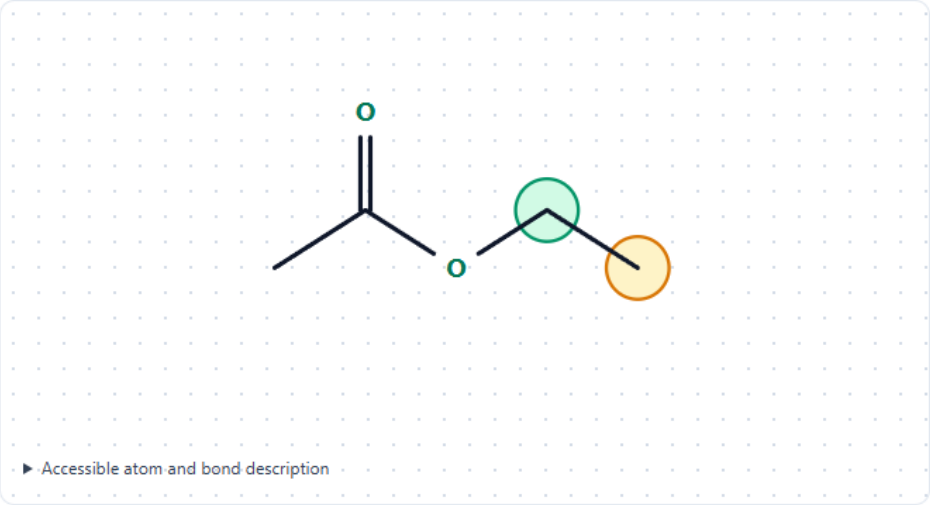
Select atoms on the structure using click or Tab and Enter. Selected: **E**

E is the methylene directly attached to the ester oxygen.

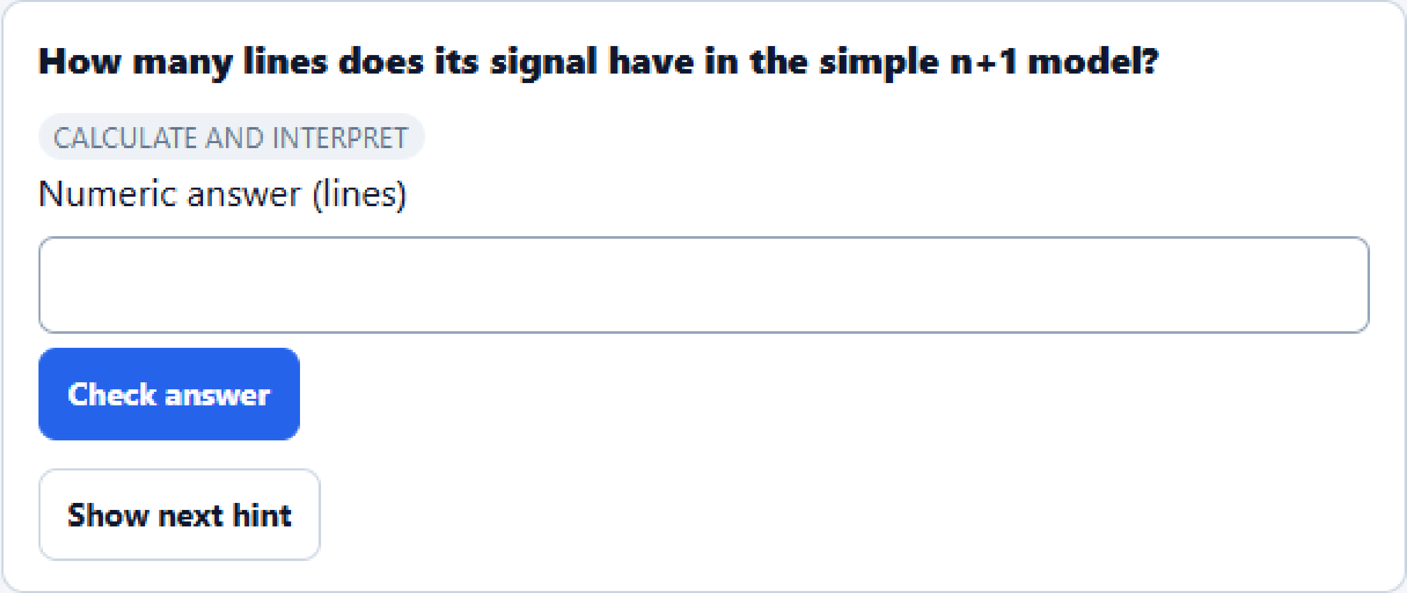
nmr-ethyl:stage-2 | How many lines does its signal have in the simple n+1 model?

Answer type: number | unit: lines

Screenshot: nmr-ethyl-stage-2-molecule


O
O
▶ Accessible atom and bond description

Screenshot: nmr-ethyl-stage-2-question



How many lines does its signal have in the simple n+1 model?

CALCULATE AND INTERPRET

Numeric answer (lines)

Check answer
Show next hint

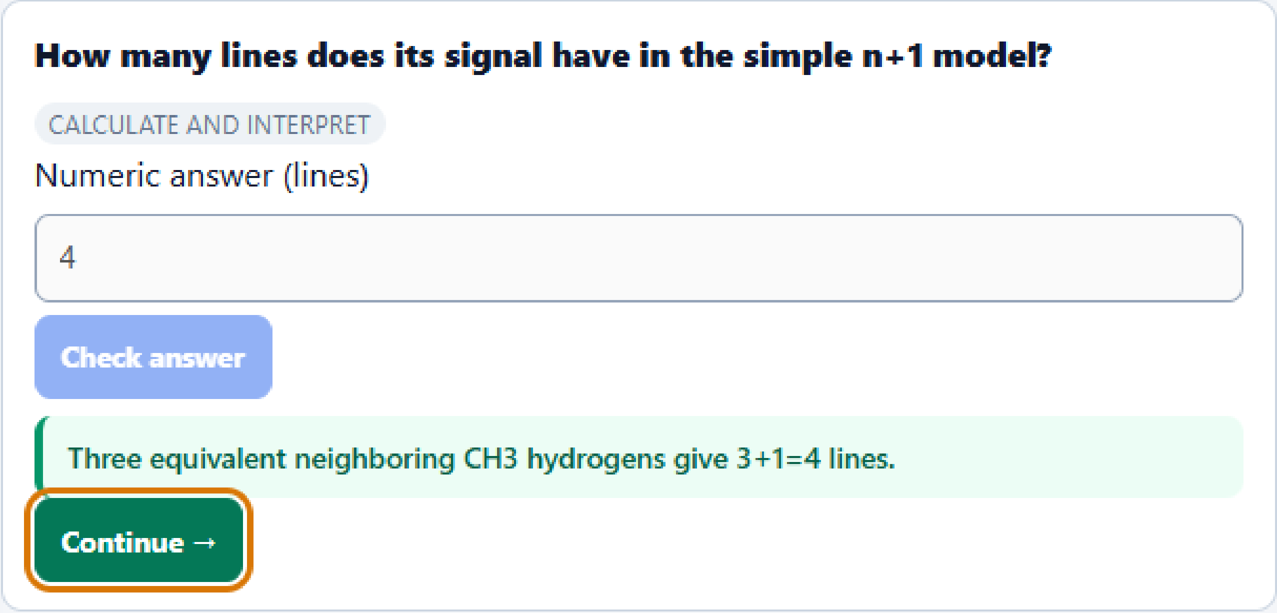
{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count hydrogens on the adjacent methyl.

Hint 2: Do not count the two H on the observed carbon itself.

Explanation: Three equivalent neighboring CH3 hydrogens give 3+1=4 lines.

Screenshot: nmr-ethyl-stage-2-feedback



How many lines does its signal have in the simple n+1 model?

CALCULATE AND INTERPRET

Numeric answer (lines)

4

Check answer

Three equivalent neighboring CH3 hydrogens give 3+1=4 lines.

Continue →

nmr-ethyl:stage-3 | The terminal ethyl CH3 gives an approximate...

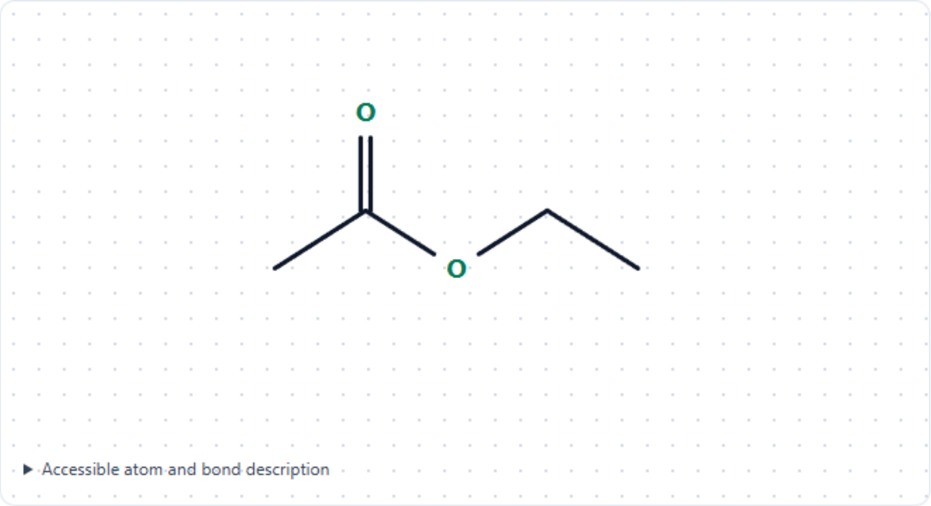
Answer type: choice

Option 0 (A): Triplet, integrating to 3H

Option 1 (B): Quartet, integrating to 2H

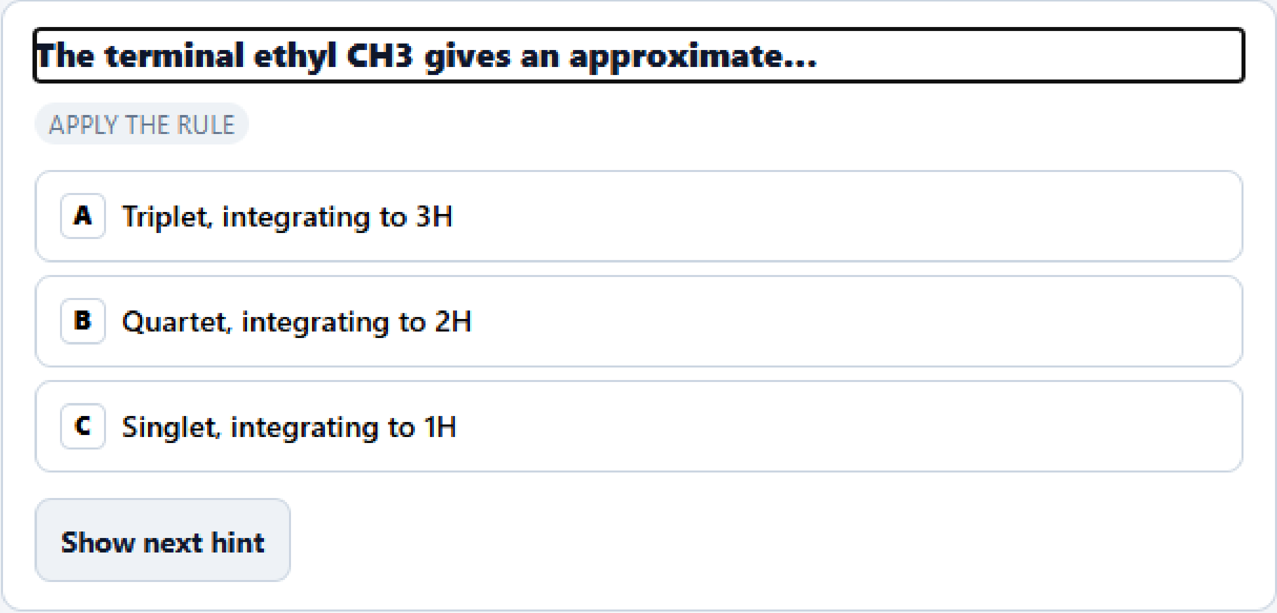
Option 2 (C): Singlet, integrating to 1H

Screenshot: nmr-ethyl-stage-3-molecule



O
O
▸ Accessible atom and bond description

Screenshot: nmr-ethyl-stage-3-question



The terminal ethyl CH3 gives an approximate...

APPLY THE RULE

A Triplet, integrating to 3H

B Quartet, integrating to 2H

C Singlet, integrating to 1H

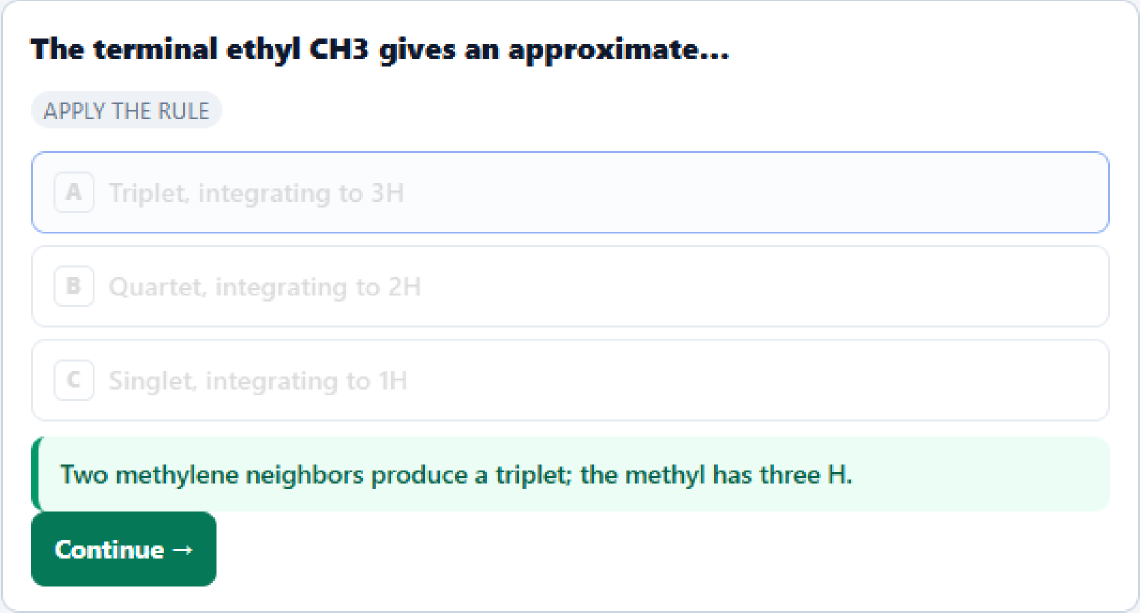
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Splitting and integration are separate measurements.

Hint 2: Use n=2 for the neighbors.

Explanation: Two methylene neighbors produce a triplet; the methyl has three H.

Screenshot: nmr-ethyl-stage-3-feedback



The terminal ethyl CH3 gives an approximate...

APPLY THE RULE

A Triplet, integrating to 3H

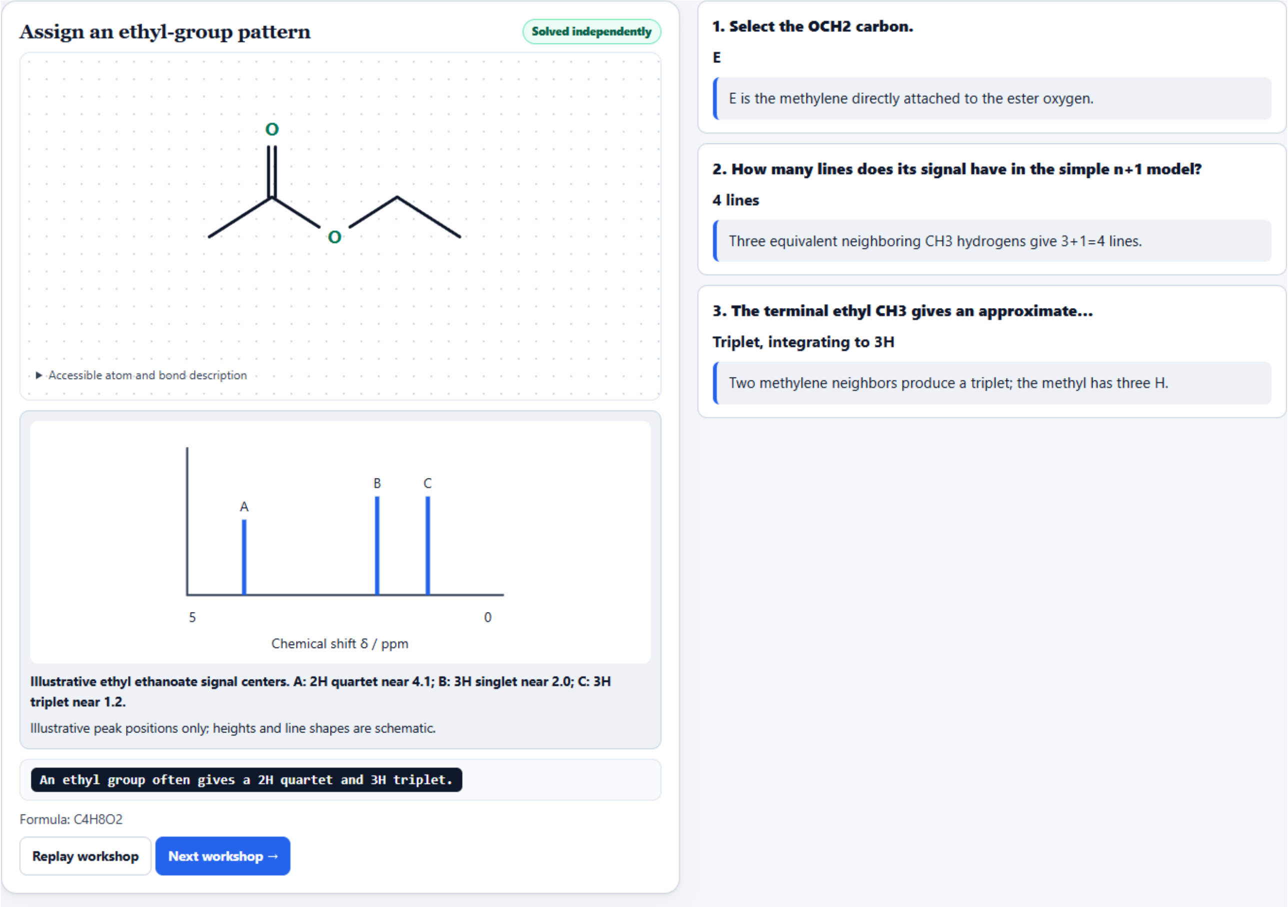
B Quartet, integrating to 2H

C Singlet, integrating to 1H

Two methylene neighbors produce a triplet; the methyl has three H.

Completion summary: An ethyl group often gives a 2H quartet and 3H triplet.

Screenshot: nmr-ethyl-complete


Solved independently
O
O
A
B
C
5
0
Chemical shift δ / ppm
▶ Accessible atom and bond description
Replay workshop
Next workshop →

Assign an ethyl-group pattern

Illustrative ethyl ethanoate signal centers. A: 2H quartet near 4.1; B: 3H singlet near 2.0; C: 3H triplet near 1.2.

Illustrative peak positions only; heights and line shapes are schematic.

`An ethyl group often gives a 2H quartet and 3H triplet.`

Formula: $C_4H_8O_2$

1. Select the OCH2 carbon.

E

> E is the methylene directly attached to the ester oxygen.

2. How many lines does its signal have in the simple n+1 model?

4 lines

> Three equivalent neighboring CH3 hydrogens give 3+1=4 lines.

3. The terminal ethyl CH3 gives an approximate...

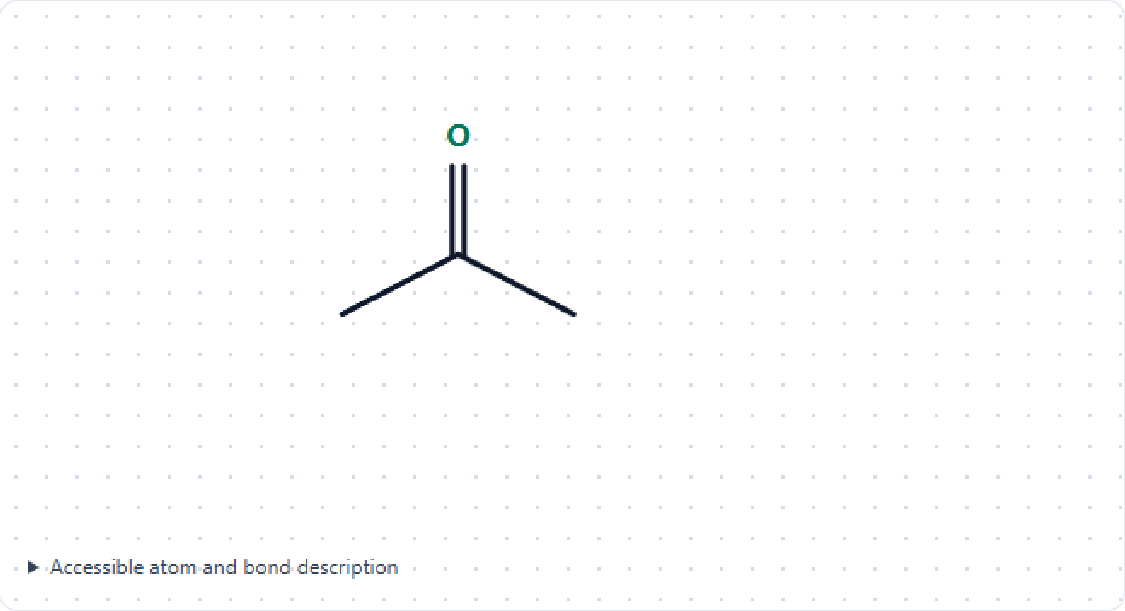
Triplet, integrating to 3H

> Two methylene neighbors produce a triplet; the methyl has three H.

combined-data | Test a structure against combined data

Supplied illustrative data: C3H6O; a ketone-like C=O band; one proton NMR singlet integrating to 6H near 2 ppm.

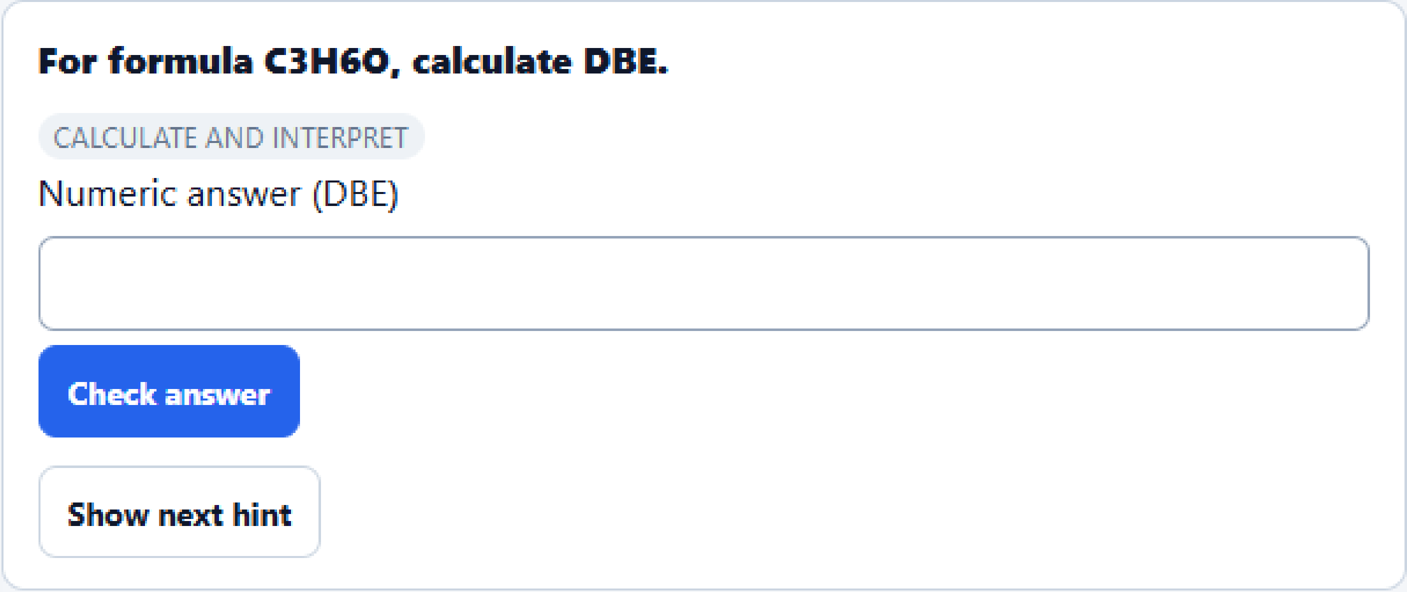
Screenshot: combined-data-stage-1-molecule



combined-data:stage-1 | For formula C3H6O, calculate DBE.

Answer type: number | unit: DBE

Screenshot: combined-data-stage-1-question



For formula C3H6O, calculate DBE.

CALCULATE AND INTERPRET

Numeric answer (DBE)

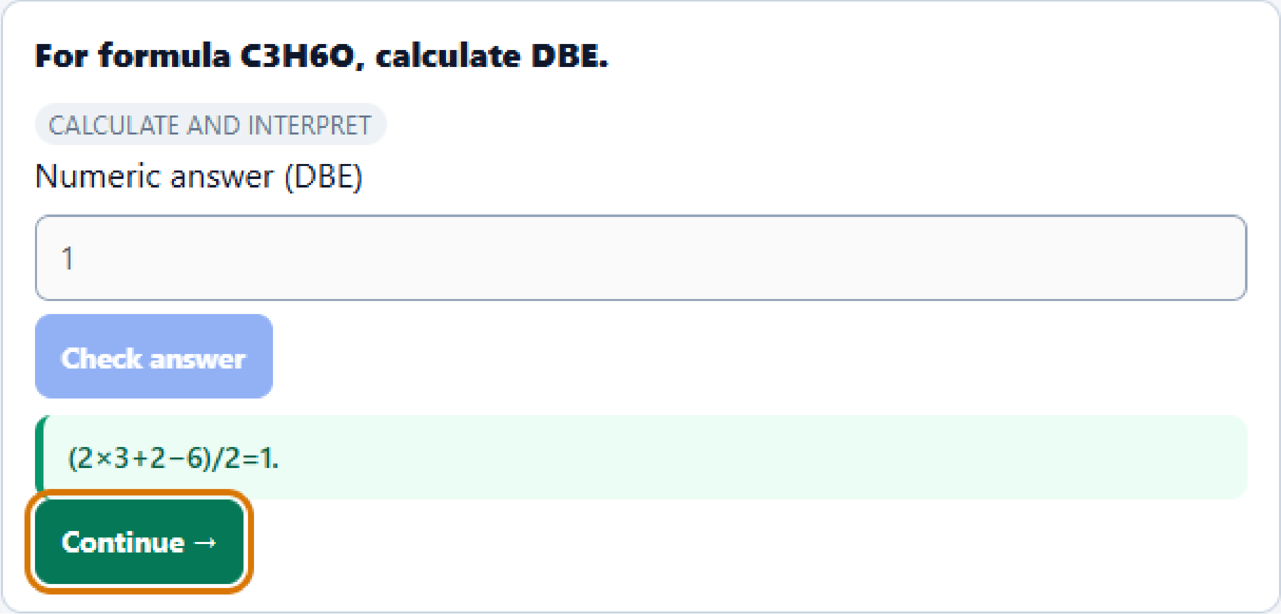
{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Ignore O in this standard formula.

Hint 2: Compare with the saturated acyclic C3H8 formula.

Explanation: (2×3+2−6)/2=1.

Screenshot: combined-data-stage-1-feedback



For formula C3H6O, calculate DBE.

CALCULATE AND INTERPRET

Numeric answer (DBE)

1

Check answer

(2×3+2−6)/2=1.

Continue →

combined-data:stage-2 | A single 6H singlet near 2 ppm suggests...

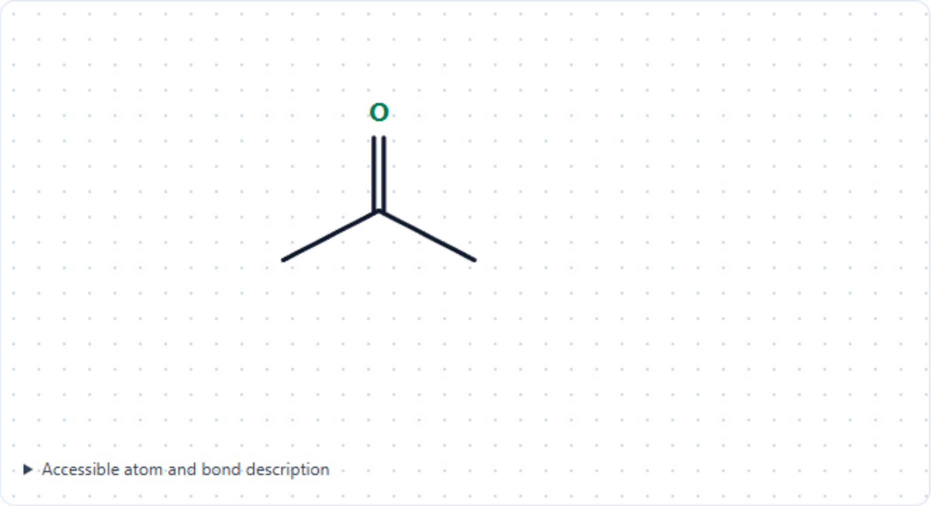
Answer type: choice

Option 0 (A): Two equivalent methyl groups with no adjacent H-bearing carbon

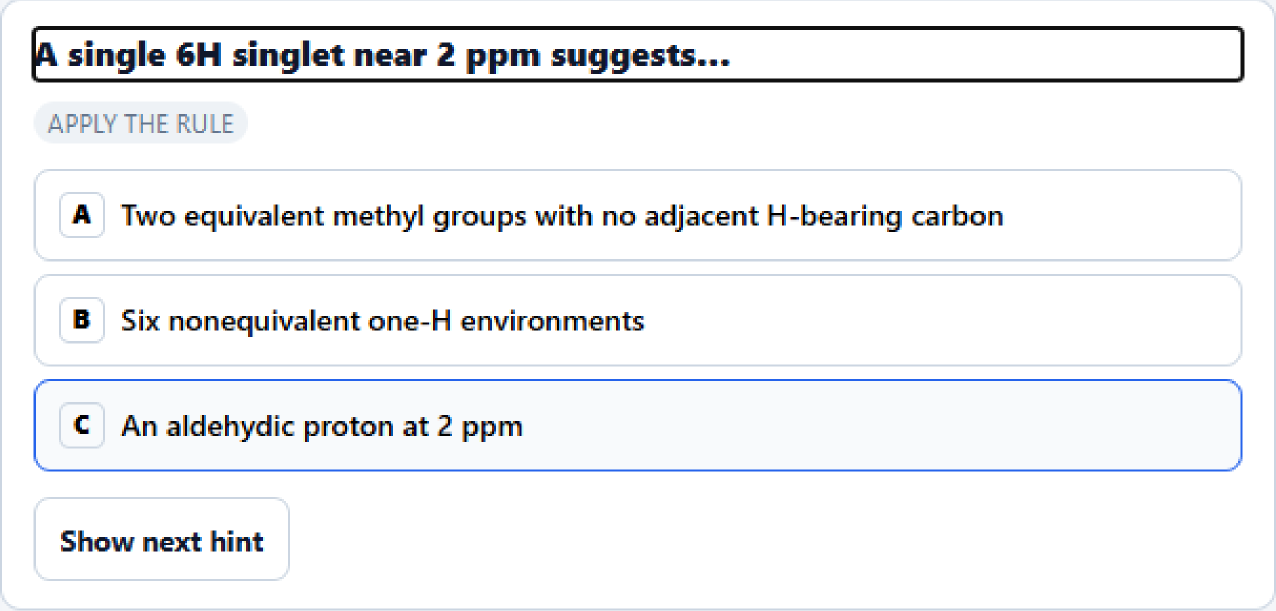
Option 1 (B): Six nonequivalent one-H environments

Option 2 (C): An aldehydic proton at 2 ppm

Screenshot: combined-data-stage-2-molecule


O
▸ Accessible atom and bond description

Screenshot: combined-data-stage-2-question


A single 6H singlet near 2 ppm suggests...

APPLY THE RULE

A Two equivalent methyl groups with no adjacent H-bearing carbon

B Six nonequivalent one-H environments

C An aldehydic proton at 2 ppm

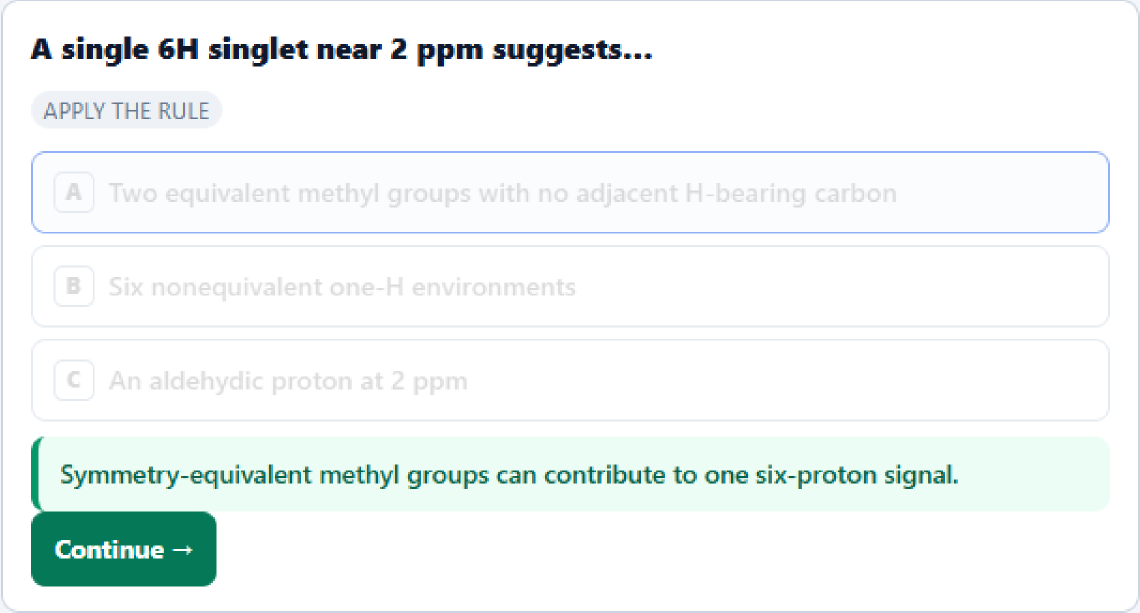
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Integration counts all protons in the environment.

Hint 2: An aldehyde H would usually be much farther downfield.

Explanation: Symmetry-equivalent methyl groups can contribute to one six-proton signal.

Screenshot: combined-data-stage-2-feedback


A single 6H singlet near 2 ppm suggests...

APPLY THE RULE

A Two equivalent methyl groups with no adjacent H-bearing carbon

B Six nonequivalent one-H environments

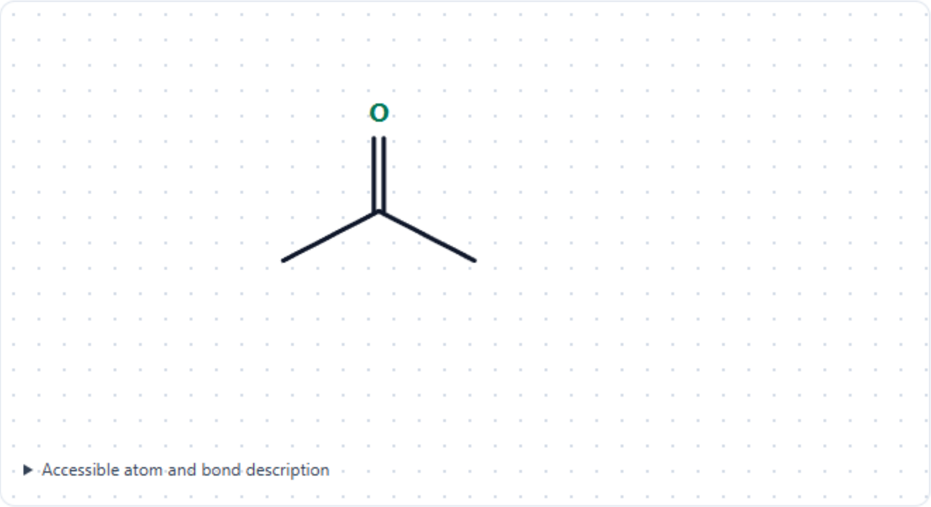
C An aldehydic proton at 2 ppm

Symmetry-equivalent methyl groups can contribute to one six-proton signal.

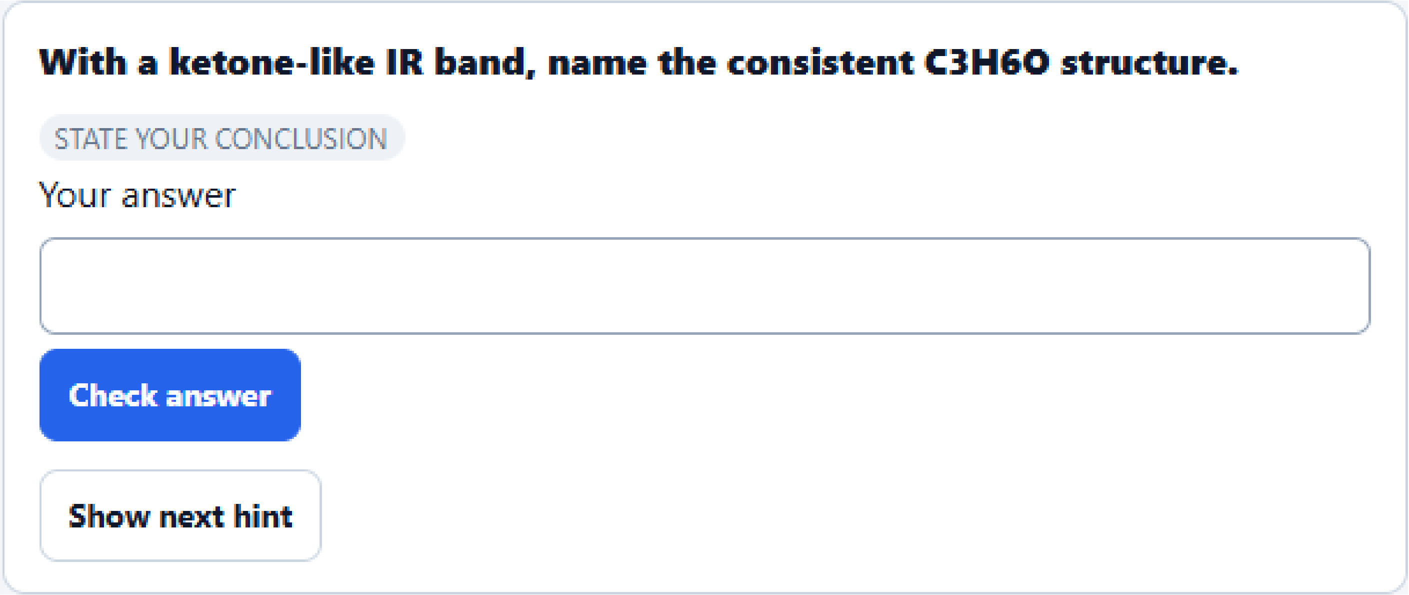
combined-data:stage-3 | With a ketone-like IR band, name the consistent C3H6O structure.

Answer type: text

Screenshot: combined-data-stage-3-molecule


O
▶ Accessible atom and bond description

Screenshot: combined-data-stage-3-question



With a ketone-like IR band, name the consistent C_3H_6O structure.

STATE YOUR CONCLUSION

Your answer

Check answer
Show next hint

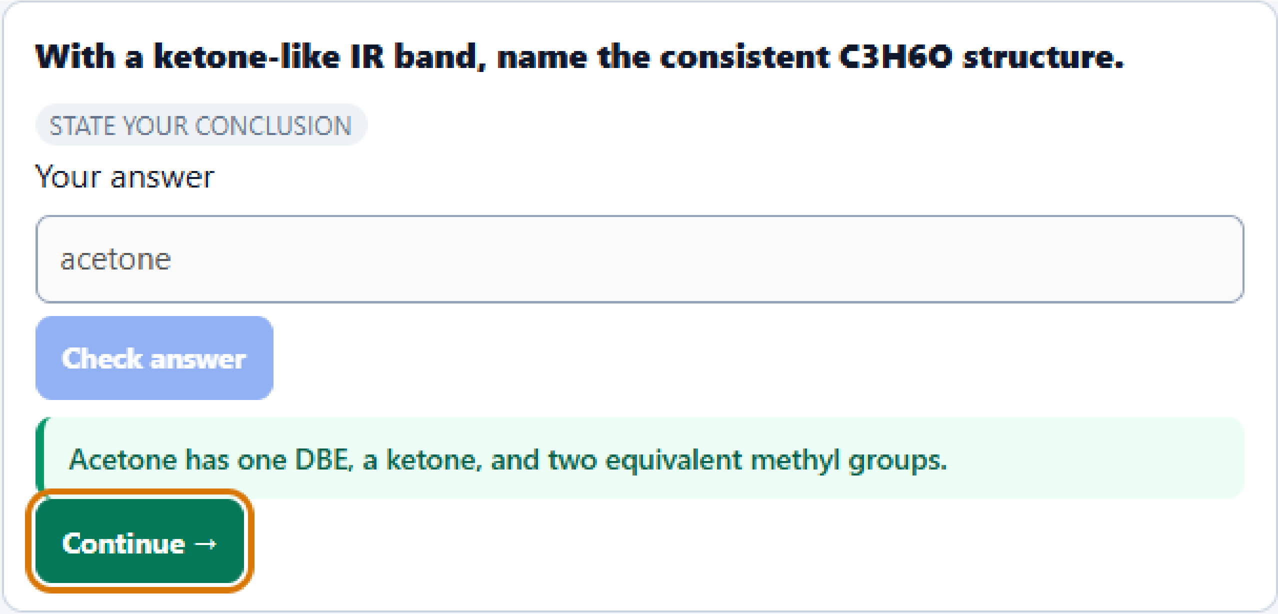
{"authorAnswer": ["acetone", "propanone", "propan-2-one"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Place C=O between two methyl groups.

Hint 2: Check all three observations together.

Explanation: Acetone has one DBE, a ketone, and two equivalent methyl groups.

Screenshot: combined-data-stage-3-feedback



With a ketone-like IR band, name the consistent C3H6O structure.

STATE YOUR CONCLUSION

Your answer

acetone

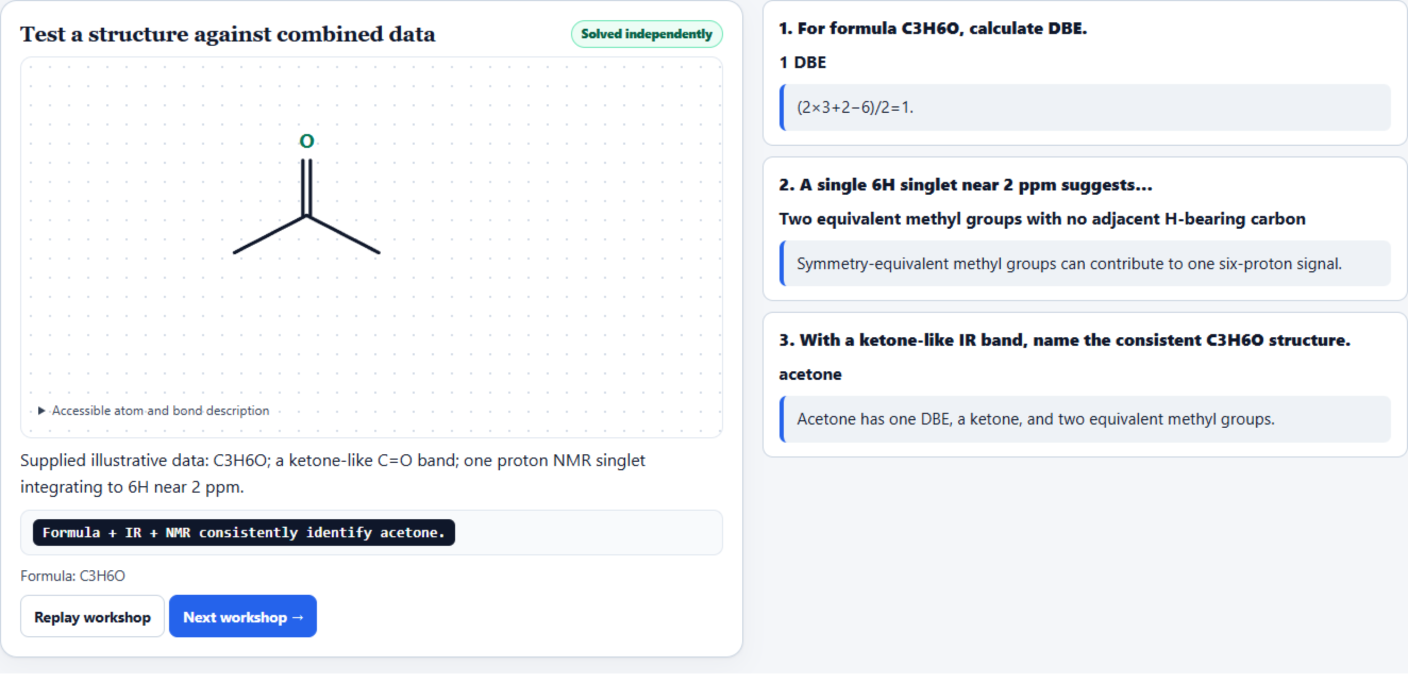
Check answer

Acetone has one DBE, a ketone, and two equivalent methyl groups.

Continue →

Completion summary: Formula + IR + NMR consistently identify acetone.

Screenshot: combined-data-complete



Test a structure against combined data

Solved independently
O
▸ Accessible atom and bond description

Supplied illustrative data: C3H6O; a ketone-like C=O band; one proton NMR singlet integrating to 6H near 2 ppm.

`Formula + IR + NMR consistently identify acetone.`

Formula: C3H6O

Replay workshop
Next workshop →

1. For formula C3H6O, calculate DBE.

1 DBE

(2×3+2−6)/2=1.

2. A single 6H singlet near 2 ppm suggests...

Two equivalent methyl groups with no adjacent H-bearing carbon

Symmetry-equivalent methyl groups can contribute to one six-proton signal.

3. With a ketone-like IR band, name the consistent C3H6O structure.

acetone

Acetone has one DBE, a ketone, and two equivalent methyl groups.

Molecular repository: author claims to verify

acetone

Author claims: propanone | formula C3H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

isopropanol

Author claims: propan-2-ol | formula C3H8O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

aceticacid

Author claims: ethanoic acid | formula C2H4O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

butadiene

Author claims: buta-1,3-diene | formula C4H6 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2]]

ethene

Author claims: ethene | formula C2H4 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2]]

ethylacetate

Author claims: ethyl ethanoate | formula C4H8O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=45.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=100.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=100.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=155.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=210.00, y=85.00; charge=0; explicit H override=none

5 (F): C; x=265.00, y=120.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1], [3, 4, 1], [4, 5, 1]]

acetaldehyde

Author claims: ethanal | formula C2H4O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

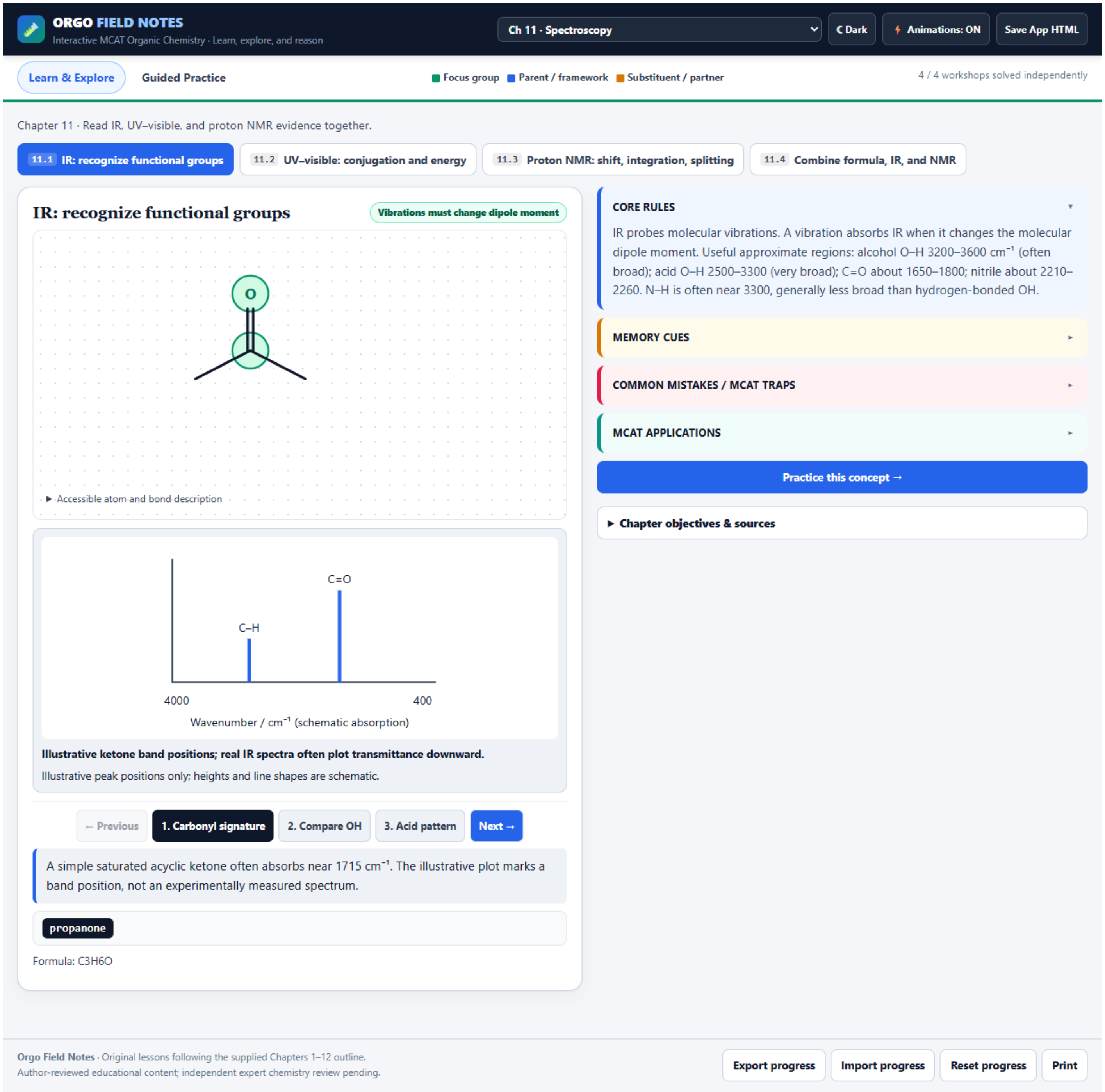
2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light


ORGO FIELD NOTES
Interactive MCAT Organic Chemistry · Learn, explore, and reason
Ch 11 · Spectroscopy
☾ Dark
Animations: ON
Save App HTML
Learn & Explore
Guided Practice
🟩 Focus group
🟦 Parent / framework
🟧 Substituent / partner
4 / 4 workshops solved independently
11.1 IR: recognize functional groups
11.2 UV–visible: conjugation and energy
11.3 Proton NMR: shift, integration, splitting
11.4 Combine formula, IR, and NMR
Vibrations must change dipole moment
O
▸ Accessible atom and bond description
C=O
C–H
4000
400
Wavenumber / cm⁻¹ (schematic absorption)
← Previous
1. Carbonyl signature
2. Compare OH
3. Acid pattern
Next →
propanone
MEMORY CUES
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MCAT APPLICATIONS
Practice this concept →
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Chapter 11 · Read IR, UV–visible, and proton NMR evidence together.

IR: recognize functional groups

Illustrative ketone band positions; real IR spectra often plot transmittance downward.

Illustrative peak positions only; heights and line shapes are schematic.

A simple saturated acyclic ketone often absorbs near 1715 cm^{-1}. The illustrative plot marks a band position, not an experimentally measured spectrum.

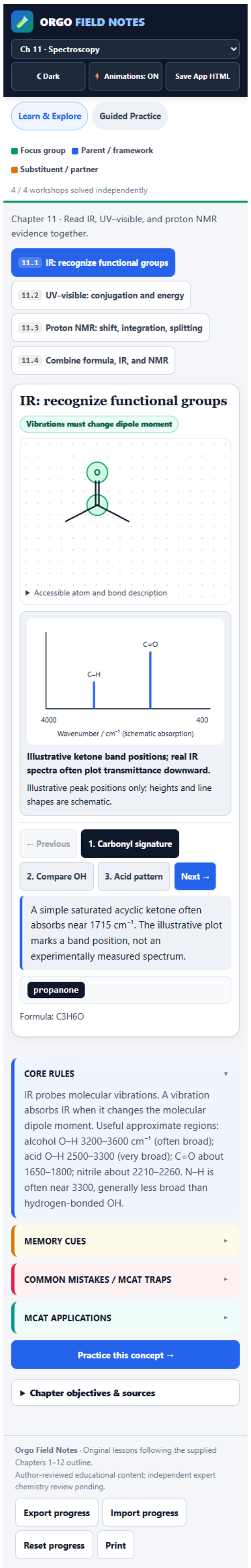
Formula: C3H6O

CORE RULES

IR probes molecular vibrations. A vibration absorbs IR when it changes the molecular dipole moment. Useful approximate regions: alcohol O–H 3200–3600 cm^{-1} (often broad); acid O–H 2500–3300 (very broad); C=O about 1650–1800; nitrile about 2210–2260. N–H is often near 3300, generally less broad than hydrogen-bonded OH.

Orgo Field Notes · Original lessons following the supplied Chapters 1–12 outline.
Author-reviewed educational content; independent expert chemistry review pending.

Screenshot: layout-mobile-light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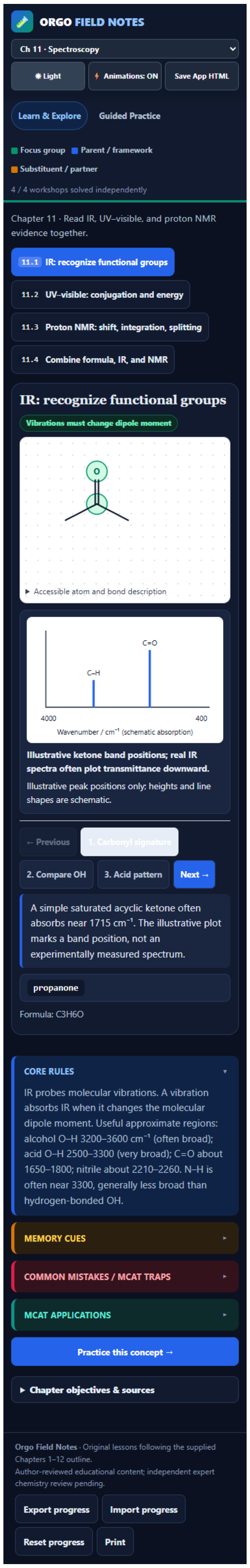
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Execution evidence

60 screenshots and 12 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.